MAMMA ADA CENTRE

In Collaboration with FACM ITALY

Class 10

AI Learning Today Leading Tomorrow

GAIO

Global AI Olympiad

Think | Explore | Create | A Better Tomorrow

IDEAS
SKILLS
SOLUTIONS
GLOBAL RECOGNITION

INNOVATION
RESEARCH
LEADERSHIP
GLOBAL RECOGNITION

AI & DATA SCIENCE
TECHNOLOGY
ENGINEERING
MATHEMATICS
CREATIVITY
GLOBAL PERSPECTIVE

1. AI Basics with Real Life
2. Communicate with AI
3. AI Applications
4. AI & Career Enhancement
5. AI Tools with Demos
6. Responsible AI & Ethics

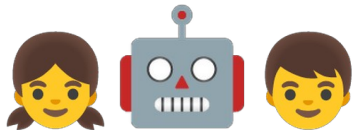
This book belongs to

Name: _____

Class & Section: _____

School: _____

📖 PREFACE



Artificial Intelligence is shaping how we learn, communicate, travel, farm, heal and create. As senior students, you are ready not only to use AI tools, but to understand how they work, question their answers and use them responsibly. This book is designed to build exactly those skills.

This book has been prepared for the Global AI Olympiad (GAIO) by Mamma Ada Centre, in collaboration with FACM, Italy. It follows the GAIO six-month journey: AI Discover, AI Connect, AI Solve, AI Rise, AI Create and AI Care. Each month introduces two topics suited to the curiosity and abilities of Class 10 learners.

Every topic opens with a clear introduction and continues with "Tech Talk", a group chat in which Riya, Kabir and their AI mentor Bolt explore ideas through real-life examples. A "How it works" flow, an India Spotlight, a hands-on Demo Lab with real AI tools, an "AI Chronicles" story, worksheets, quizzes, and standard and case-based exercises help you learn by reading, doing, questioning and creating.

Values are at the heart of this book. Along with learning what AI can do, children learn that people create and guide technology, that AI can make mistakes, and that safety, kindness and privacy matter online and offline.

🤝 A note for teachers and parents

Encourage students to try the Demo Labs under supervision, discuss the case studies, and question AI answers rather than accept them blindly. Praise reasoning and curiosity as much as right answers. An answer key is provided at the end of the book.

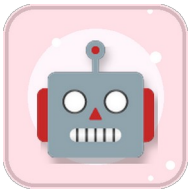
We hope this book sparks curiosity, creativity and confidence in every young learner. Curious minds, brighter futures!

With best wishes,

The GAIO Academic Team

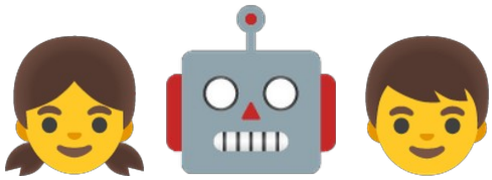
Mamma Ada Centre • In collaboration with FACM, Italy

Welcome, Seniors!



MEET BOLT!

Welcome, Class 10! I'm Bolt. This year: how models learn, how GenAI works, testing code, design thinking, your AI career road and responsible AI. Let's lead!



Look for these pictures in your book:

TECH TALK Concepts in a group chat

AI CHRONICLES Stories of teens & AI

LOOK AROUND YOU See AI in real life

LET'S DO IT! Do a fun activity

WORKSHEET Write, tick and draw

PUZZLE FUN Solve puzzles

DEMO LAB Hands-on AI tool demo

QUIZ TIME Answer quiz questions

TRUE OR FALSE True or false?

EXERCISE Standard & case-based Qs

MAGIC WORDS New words to learn

UNIT REVIEW Check your unit

My 6-Month AI Journey

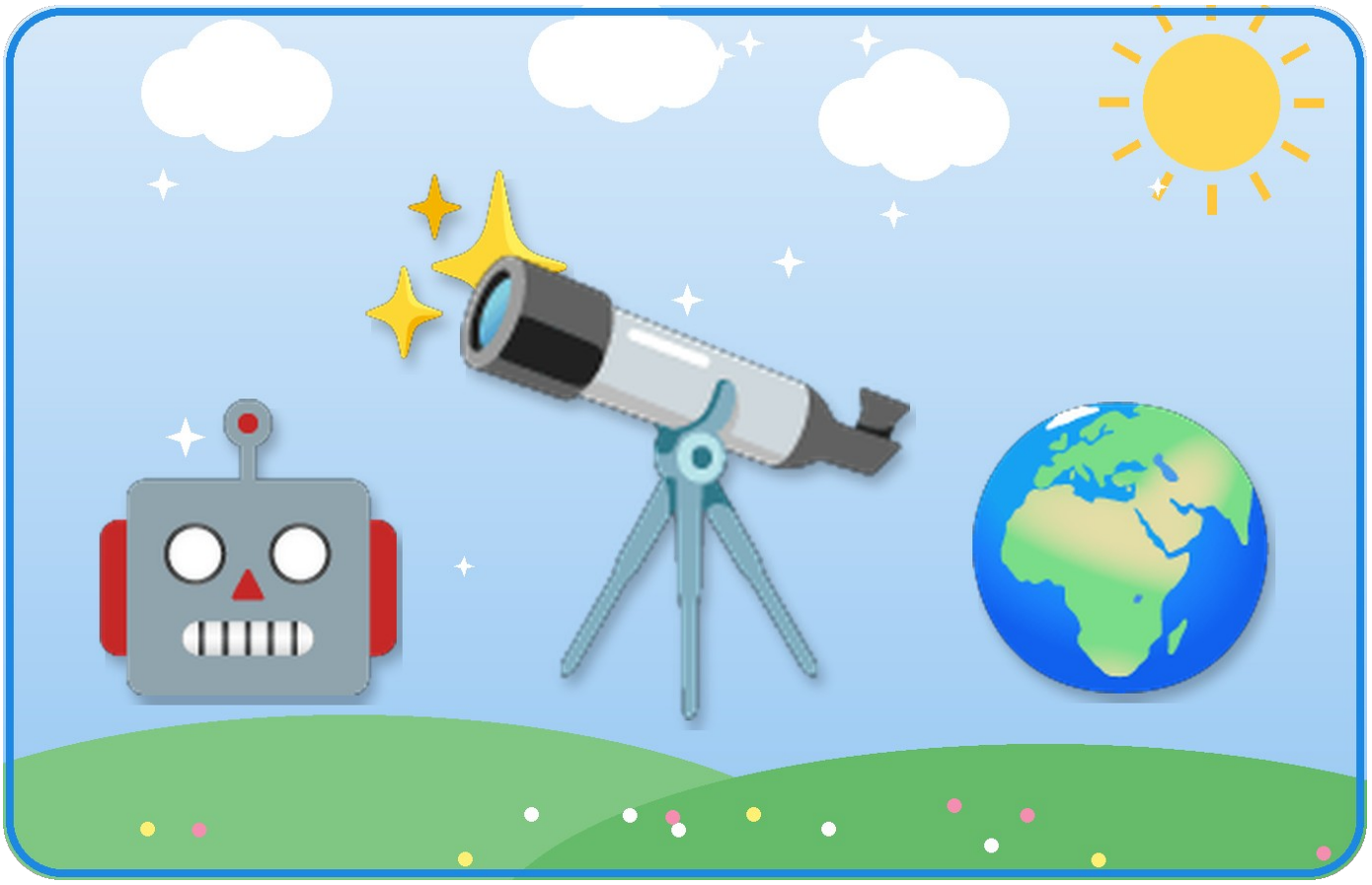
Follow the path. One month at a time!

MONTH 1 **AI DISCOVER** AI Basics	**How AI Thinks with Data • Generative AI Uncovered**
MONTH 2 **AI CONNECT** Communicate with AI	**AI-Assisted Coding • Code → Test → Improve**
MONTH 3 **AI SOLVE** AI Applications	**AI at Work • AI Solving Real Problems**
MONTH 4 **AI RISE** AI & Career	**Your Road to an AI Career • AI Skills Beyond School**
MONTH 5 **AI CREATE** AI Tools	**AI Productivity Booster • Create Your AI Project**
MONTH 6 **AI CARE** Responsible AI	**Original or AI-Made? • Use AI, Don't Misuse AI**

MONTH 1

AI DISCOVER

See • Understand • Explore



Area: AI Basics

In this unit, you will explore:

Topic 1

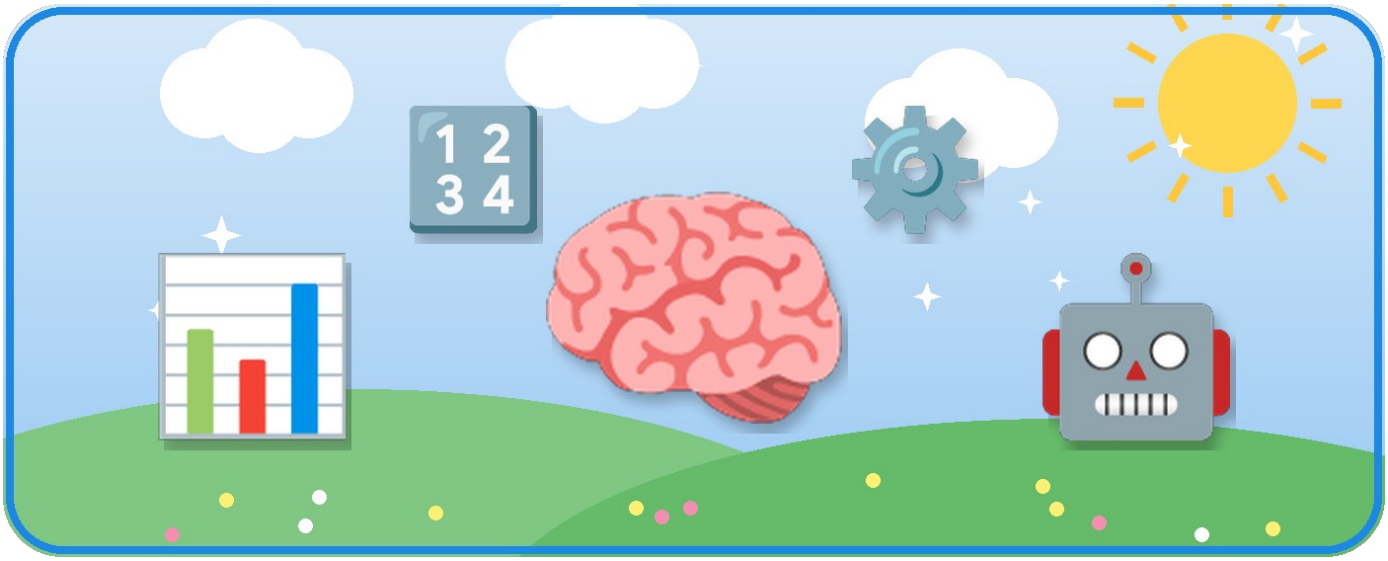
How AI Thinks with Data

Topic 2

Generative AI Uncovered

TOPIC 1 • MONTH 1 • AI BASICS

How AI Thinks with Data



LET'S LEARN

- AI sees data as numbers called features.
- It learns weights: how much each feature matters.
- Too much memorising = overfitting. Balance is key!

Inside an AI model

Features

Weights

Labels

Prediction

MAGIC WORDS

Feature An input property

Weight Importance of a feature

Overfitting Memorising, not learning

💬 TECH TALK: How AI Thinks with Data

💬 GAIO Squad Riya, Kabir, Bolt • online

👦 Kabir

Bolt, how does AI 'see' a house to predict its price 🏠?

10:02

🤖 Bolt

As numbers called features 🔢: area, rooms, distance to metro, age of building.

10:04 ✓✓

👧 Riya

Does every feature matter equally?

10:06

🤖 Bolt

No. The model learns weights ⚖️: area may matter a lot, paint colour hardly at all.

10:08 ✓✓

👦 Kabir

How does it learn the weights?

10:10

🤖 Bolt

It guesses, checks the error, then adjusts weights again and again to reduce it 🔁.

10:12 ✓✓

👧 **Riya**

What's overfitting?

10:14

🤖 **Bolt**

When a model memorises training data, noise and all, then fails on new data 📉. Like mugging up answers without understanding!

10:16 ✓✓

👦 **Kabir**

So we want a balanced model that generalises 🎯.

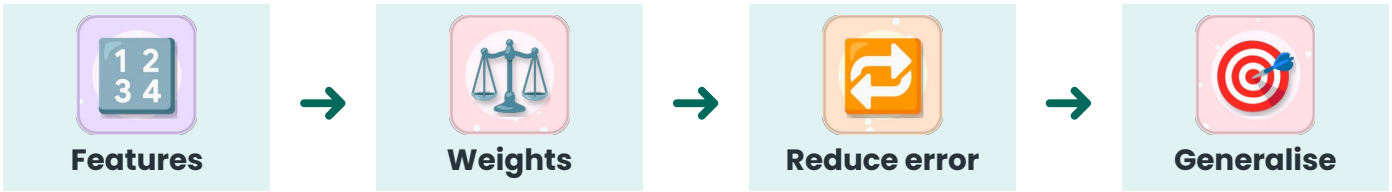
10:18

😊 Type a message...

📌 **Pinned by Bolt**

AI turns data into features, learns weights for how much each matters and reduces error step by step. Overfitting means memorising instead of generalising.

🔄 How it works



INDIA SPOTLIGHT

Indian property and agri-tech start-ups use feature-based ML models to estimate prices and crop yields.

AI CHRONICLES: The Model That Memorised

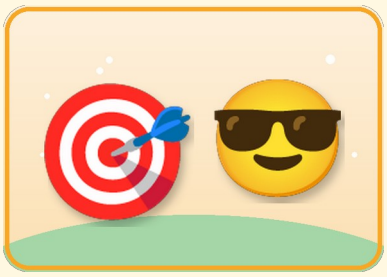
Riya Kabir Mr. Rao

SCENE 1

For the science fair, Riya and Kabir built a model to predict house rents in their town 💰.

Features: area, rooms, floor, metro distance. The model learned weights through many rounds.

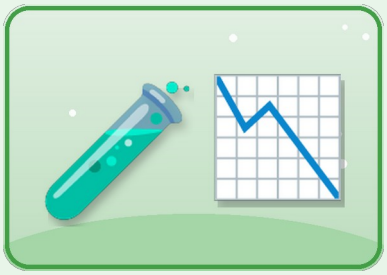
SCENE 2



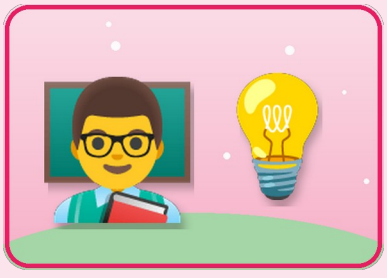
SCENE 3

On training data it was 99% accurate! They were sure they'd win 🏆.

Mr. Rao tested it on 20 new houses. Accuracy crashed to 55%. The judges frowned.



SCENE 4



SCENE 5

"It memorised your data, including odd one-off rents. That's overfitting," Mr. Rao explained.

They simplified the model and used more varied data. New-house accuracy rose to 84% 🎉.

SCENE 6

🔮 What happens next? Tick ✓ one!

☐	☐	☐
They add a 'season' feature and retest	**They test only on training data**	**The houses start predicting rent**

🧐 Story check: answer the questions.

1. Which features did the model use?

2. Why did accuracy drop on new houses?

3. How did they fix overfitting?

Rate this story: shade the stars.

☆ ☆ ☆ ☆ ☆

LOOK AROUND YOU: Features AI looks at

House: size, rooms, area

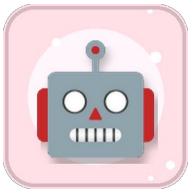
Email: words, links, sender

Health: age, BP, sugar

Video: watch time, likes

Crop: rain, soil, heat

Car: speed, distance, signal



BOLT SAYS:

A student who memorises answers fails new questions. So does an overfitted AI!

FUN FACT!

To a computer, a photo is just a grid of numbers: one for each pixel's colour!

I spy AI! Draw one AI helper you saw today.

LET'S DO IT! Human Neural Network

You need:

Feature cards

Weight tokens

Groups of 5

Steps:

1		3 students hold features: rain, sun, soil.
2		Each gets a weight: 1, 2 or 3 tokens.
3		The 4th student adds feature × weight.
4		The 5th decides: good harvest or not?

THINK & TALK

What happens if you change a weight?

Draw what you did!

WORKSHEET: How AI Thinks with Data (Part A)

Name: _____ Date: _____

1. Match the AI term to its meaning.

Feature ●

● **Importance of input**

Weight ●

● **Memorising data**

Label ●

● **Input property**

Overfitting ●

● **Correct answer tag**

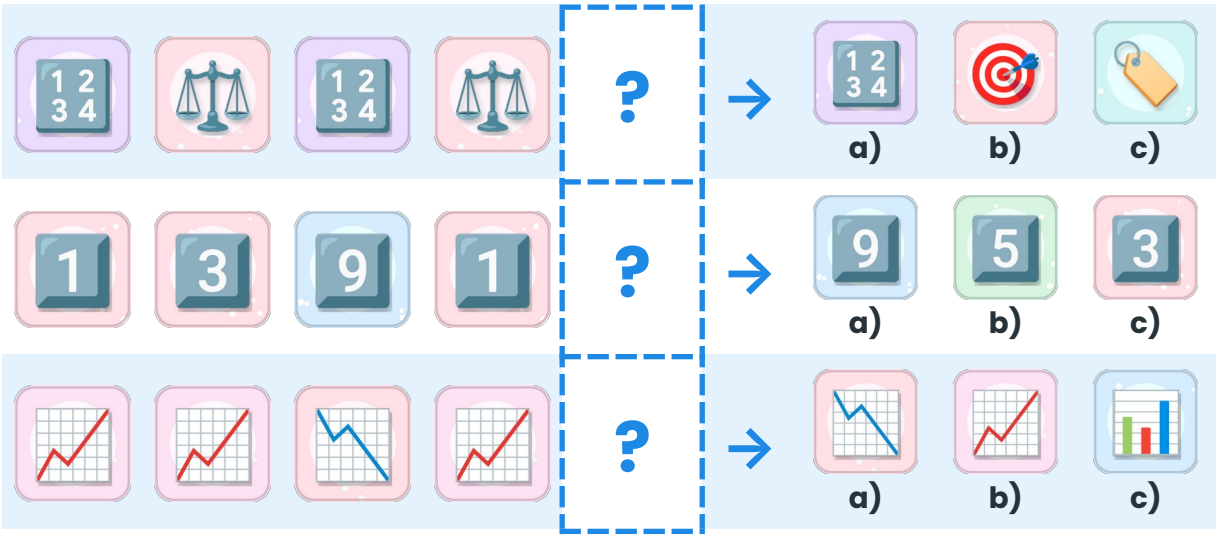
2. Feature or label? (Predicting house price) Tick ✓.

	Feature	Label
Size in sq ft		
Number of rooms		
House price		
Location		

WORKSHEET: How AI Thinks with Data (Part B)

Name: _____ Date: _____

1. Feature patterns! Circle what comes next.



2. Fill in the blanks. Use the word box.

WORD BOX: weight ✿ feature ✿ overfitting

An input property is a _____.

How much it matters is its _____.

Memorising, not learning, is _____.

3. Bonus: Draw your favourite picture from this topic!

I drew _____

PUZZLE FUN: How AI Thinks with Data

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.

A				
B				
C				

2. Define it! Write each term in your own words with an example.

Feature

Meaning: _____

Example: _____

Weight

Meaning: _____

Example: _____

Overfitting

Meaning: _____

Example: _____

DEMO LAB: Spreadsheet (Google Sheets / Excel)

Name: _____ Date: _____

Goal: See how features and weights produce a prediction, and spot overfitting.

Open: sheets.google.com or Excel | **With a teacher or parent**

Steps

1. Teacher gives 10 houses with area, rooms and price.
2. Make a formula: price = w1 × area + w2 × rooms. Start with w1 = 3000, w2 = 50000.
3. Add an error column: actual - predicted.
4. Adjust w1 and w2 to make the total error smaller. Try 5 times.
5. Test your best weights on 3 NEW houses.

Record your observations

Best w1 and w2:	
Total error (training):	
Error on new houses:	

Reflect: If your weights were perfect on training houses but bad on new ones, what happened?

? QUIZ TIME: How AI Thinks with Data

Circle the right answer!

Q1. Inputs AI uses are called...

a) Features

b) Balloons

c) Biscuits

Q2. How much a feature matters is its...

a) Colour

b) Weight

c) Tune

Q3. Memorising instead of learning is...

a) Accuracy

b) Running

c) Overfitting

Q4. To a computer, a photo is...

a) Numbers

b) Paint

c) Magic

Q5. The correct answer tag is a...

a) Gift

b) Label

c) Sock

MY SCORE: _____ / 5. Shade one star per correct answer.

☆ ☆ ☆ ☆ ☆

✅ TRUE OR FALSE: How AI Thinks with Data

Look at the picture. Tick ✓ TRUE or FALSE.

	TRUE	FALSE
AI sees data as numbers.		
Weights show how much a feature matters.		
Overfitting means learning really well.		
Photos are grids of pixel numbers.		
A label is the correct answer tag.		
AI picks weights by rolling dice.		

HOME CONNECT

Pick a family decision (e.g., buying a phone). List the features you'd compare and their weights!

🔢 Self-assessment: tick one.

I know it well!

I know some.

I need practice.

EXERCISE: How AI Thinks with Data

Name: _____ Date: _____

A. Tick (✓) the correct answer.

1. Measurable inputs AI uses are ____.

(a) features ☐ **(b)** feelings ☐ **(c)** folders ☐

2. How much a feature matters is its ____.

(a) width ☐ **(b)** weight ☐ **(c)** warning ☐

3. Memorising training data and failing on new data is ____.

(a) underground ☐ **(b)** generalising ☐ **(c)** overfitting ☐

4. A good model ____ to new data.

(a) generalises ☐ **(b)** memorises ☐ **(c)** refuses ☐

B. Fill in the blanks. Use the word box.

error **weight** **new** **overfitting**

1. Models reduce ____ step by step.

2. Paint colour has a small ____.

3. Test on ____ data to catch overfitting.

4. Simpler models with varied data reduce ____.

C. Write T for True and F for False.

1. Features are numbers describing data. [ ]

2. All features always matter equally. [ ]

3. Overfitting looks great on training data. [ ]

4. Overfitting improves new-data accuracy. [ ]

D. Match the following. Write the letter.

Column A		Column B
1. Feature	____	**(a)** Memorising noise
2. Weight	____	**(b)** Input variable
3. Overfitting	____	**(c)** Work on new data
4. Generalise	____	**(d)** Importance of a feature

E. Answer in one word.

1. What are AI's input variables? _____

2. What shows a feature's importance? _____

3. What is memorising instead of learning? _____

F. Answer these questions.

1. Explain features and weights using a house-price example.

2. How can we detect and reduce overfitting?

G. Case-based questions. Read and answer.

A rent model scored 99% on training data but 55% on 20 new houses. Using a simpler model and more varied data raised new-house accuracy to 84%.

1. The 99% vs 55% gap shows ____.

(a) overfitting ☐ **(b)** perfect learning ☐ **(c)** a power cut ☐

2. The 20 new houses acted as a ____.

(a) training set ☐ **(b)** test set ☐ **(c)** feature ☐

3. Accuracy improved because ____.

(a) houses changed ☐ **(b)** they cheated ☐ **(c)** the model generalised better ☐

H. THINK DEEPER

Design features for an AI that predicts school bus delays. Which features would get the highest weights, and why?

TOPIC 2 • MONTH 1 • AI BASICS

Generative AI Uncovered

LET'S LEARN

- Chatbots run on Large Language Models (LLMs).
- LLMs predict the next word, one token at a time.
- Image AI turns noise into pictures step by step!

How GenAI works

Tokens

Next-word guess

Noise → image

Hallucinations

MAGIC WORDS

Token A word piece

LLM Large Language Model

Hallucination Confident wrong answer

💬 TECH TALK: Generative AI Uncovered

💬 GAIO Squad Riya, Kabir, Bolt • online

👧 Riya

Bolt, how does a chatbot write whole paragraphs 💬?

10:02

🤖 Bolt

It runs on a Large Language Model, an LLM 🧠, trained on huge amounts of text.

10:04 ✓✓

👦 Kabir

Does it understand like us?

10:06

🤖 Bolt

It predicts the next token 🧩, a word or word-piece, again and again, based on patterns.

10:08 ✓✓

👧 Riya

And image generators 🖼️?

10:10

🤖 Bolt

Many start from random noise and remove it step by step until a picture matching the prompt appears.

10:12 ✓✓

Kabir
Why does it sometimes say false things?
10:14

Bolt
Hallucination ⚠️: it produces likely-sounding text, not checked facts. Always verify.
10:16 ✓✓

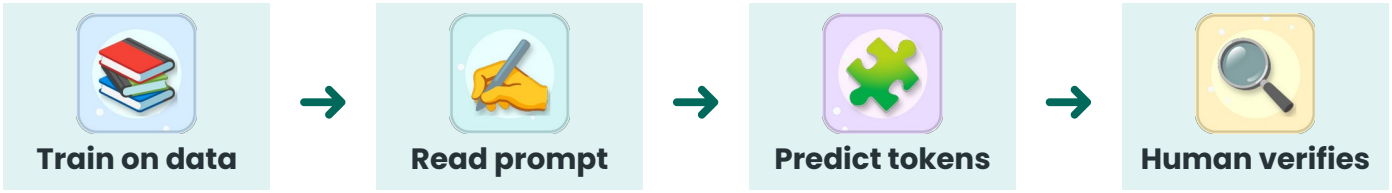
Riya
Powerful predictor, not a fact machine ✅!
10:18

😊 Type a message...

📌 **Pinned by Bolt**

LLMs generate text by predicting the next token. Image models turn noise into pictures. GenAI can hallucinate, so outputs must be verified.

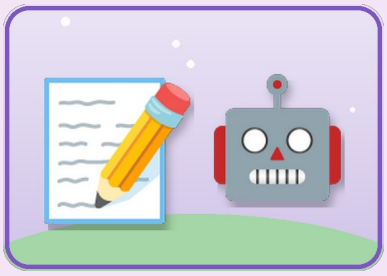
🔄 How it works



INDIA SPOTLIGHT
Indian teams are building LLMs trained on Indian languages so GenAI can serve users in their mother tongues.

AI CHRONICLES: The Confident Chatbot

Kabir | Ms. D'Souza | LLM



SCENE 1

Kabir used a chatbot to draft his history essay on India's freedom struggle 🇮🇳.

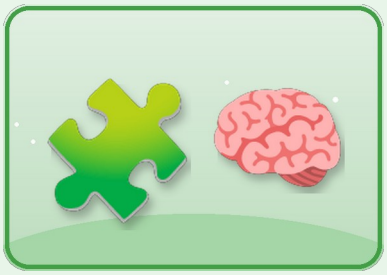
It wrote beautifully, even quoting a 'famous 1931 speech' with dates and places.

SCENE 2

SCENE 3

Ms. D'Souza asked, "Where is this speech from?" Kabir searched. It didn't exist 😳.

She explained: "An LLM predicts likely next tokens. It can create convincing fiction: hallucination."



SCENE 4

SCENE 5

Kabir rewrote the essay using textbooks and archives, using AI only to improve structure.

He got an A and a lesson: "AI writes fluently. I must write truthfully."

SCENE 6

🔮 What happens next? Tick ✓ one!

☐	☐	☐
Kabir teaches classmates to fact-check AI	**Kabir submits AI text unchecked**	**The chatbot writes its own exam**

🧐 Story check: answer the questions.

1. What did the chatbot invent?

2. Why do LLMs hallucinate?

3. How did Kabir use AI the second time?

Rate this story: shade the stars.

☆ ☆ ☆ ☆ ☆

LOOK AROUND YOU: GenAI in daily life

Chat assistants

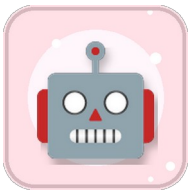
Image generators

Music makers

Code helpers

Voice assistants

Translators



BOLT SAYS:

I don't "know" things like you do. I predict likely words. That's why I can be wrong!

FUN FACT!

LLMs learn from huge amounts of text, so they can write in many styles and languages!

I spy AI! Draw one AI helper you saw today.

LET'S DO IT! Human LLM Game

You need:

Sentence strips

Circle of students

Timer

Steps:

1		First student says: "Once upon a..."
2		Next student predicts the most likely next word.
3		Continue around the circle for 20 words.
4		Did the story drift or invent things?

THINK & TALK

How is this like an LLM? How is it different?

Draw what you did!

✏️ WORKSHEET: Generative AI Uncovered (Part A)

Name: _____ Date: _____

1. Predict the next token! Use the word box.

WORD BOX: day ✿ morning ✿ tea

Good _____

A cup of _____

Happy birth_____

2. True about GenAI? Tick ✓.

	True	False
LLMs predict next words		
LLMs truly understand like humans		
GenAI can hallucinate		
GenAI is always correct		

WORKSHEET: Generative AI Uncovered (Part B)

Name: _____ Date: _____

1. Match the GenAI tool to what it makes.

Chatbot ●

Image AI ●

Music AI ●

Code AI ●

●

Tunes

●

Programs

●

Text answers

●

Pictures

2. Fill in the blanks. Use the word box.

WORD BOX: token ✿ LLM ✿ hallucination

A Large Language Model is an _____.

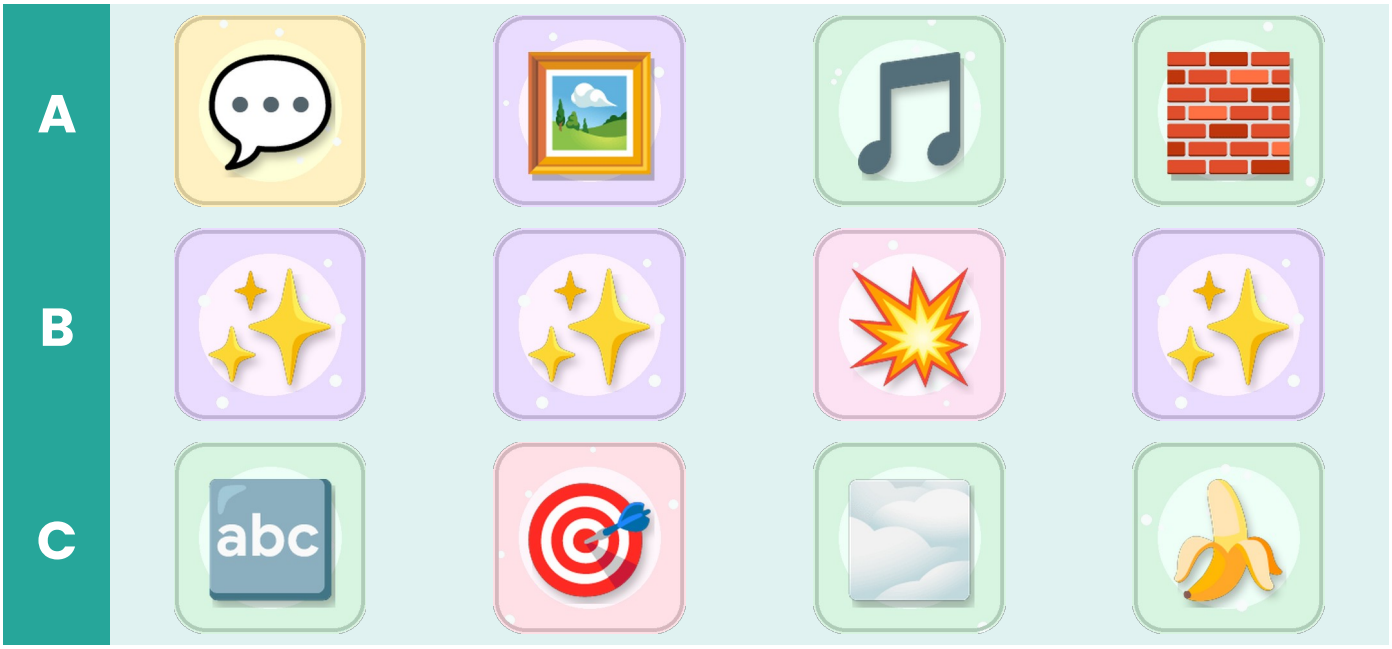
A word piece is a _____.

A confident wrong answer is a _____.

PUZZLE FUN: Generative AI Uncovered

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.



2. Define it! Write each term in your own words with an example.

Token

Meaning: _____

Example: _____

LLM

Meaning: _____

Example: _____

Hallucination

Meaning: _____

Example: _____

🧪 DEMO LAB: A school-approved chatbot + tokenizer demo (with teacher)

Name: _____ Date: _____

🎯 **Goal:** Peek inside how LLMs work and test for hallucinations.

🔗 **Open:** any approved chatbot / online tokenizer | 👩‍🏫 **With a teacher or parent**

👣 Steps

1		Paste a sentence into a tokenizer tool. Count the tokens.
2		Ask the chatbot to continue: 'The Taj Mahal is...'. Regenerate 3 times.
3		Compare the 3 answers. Why are they different?
4		Ask about a made-up person, e.g., 'Who is Dr. Zorvan Kapil?'
5		Check facts from its answers against a reliable source.

🔬 Record your observations

Token count:	
How answers differed:	
Did it hallucinate?	

💭 **Reflect:** Why does an LLM sound confident even when it is wrong?

? QUIZ TIME: Generative AI Uncovered

Circle the right answer!

Q1. LLM stands for...

a) Large Language Model

b) Lemon Lime Mix

c) Long Level Map

Q2. LLMs predict the next...

a) Weather

b) Token

c) Dice roll

Q3. A confident wrong answer is a...

a) Fact

b) Prize

c) Hallucination

Q4. Image AI starts from...

a) Noise

b) Pizza

c) Paint tubes

Q5. GenAI answers must be...

a) Trusted blindly

b) Checked

c) Ignored

MY SCORE: ____ / 5. Shade one star per correct answer.

☆ ☆ ☆ ☆ ☆

✅ TRUE OR FALSE: Generative AI Uncovered

Look at the picture. Tick ✓ TRUE or FALSE.

		👍 TRUE	👎 FALSE
	Chatbots run on LLMs.		
	LLMs predict the next token.		
	LLMs understand exactly like humans.		
	GenAI can hallucinate.		
	Image AI turns noise into pictures.		
	GenAI answers never need checking.		

HOME CONNECT

Ask an AI one question with your family. Then verify the answer from a trusted source.

🔢 Self-assessment: tick one.

I know it well!

I know some.

I need practice.

EXERCISE: Generative AI Uncovered

Name: _____ Date: _____

A. Tick (✓) the correct answer.

1. Chatbots run on _____.

(a) LLMs ☐ **(b)** LEDs ☐ **(c)** LANs ☐

2. LLMs predict the next _____.

(a) number plate ☐ **(b)** token ☐ **(c)** photo ☐

3. Confident but false AI output is _____.

(a) citation ☐ **(b)** compression ☐ **(c)** hallucination ☐

4. Many image generators start from random _____.

(a) noise ☐ **(b)** music ☐ **(c)** maps ☐

B. Fill in the blanks. Use the word box.

Language piece verified patterns

1. LLM stands for Large _____ Model.

2. A token can be a word or word-_____.

3. GenAI output must be _____.

4. LLMs learn _____ from huge text data.

C. Write T for True and F for False.

1. LLMs generate text token by token. [ ]

2. LLMs always check facts before answering. [ ]

3. Regenerating can give different answers. [ ]

4. Hallucinations never sound confident. [ ]

D. Match the following. Write the letter.

Column A		Column B
1. LLM	____	**(a)** Invented output
2. Token	____	**(b)** Large Language Model
3. Hallucination	____	**(c)** Input instruction
4. Prompt	____	**(d)** Word or word-piece

E. Answer in one word.

1. What does an LLM predict next? _____

2. What is AI inventing facts called? _____

3. What starts many image generations? _____

F. Answer these questions.

1. Explain in simple terms how an LLM writes a sentence.

2. Why must students verify GenAI output?

G. Case-based questions. Read and answer.

A chatbot wrote Kabir's essay with a quote from a '1931 speech' that never existed. He rewrote it using textbooks, using AI only to improve structure.

1. The fake quote was a ____.

(a) hallucination ☐ **(b)** citation ☐ **(c)** token ☐

2. The chatbot was predicting ____.

(a) facts only ☐ **(b)** likely next tokens ☐ **(c)** the future ☐

3. Kabir's second approach was ____.

(a) copying ☐ **(b)** deleting history ☐ **(c)** responsible use ☐

H. THINK DEEPER

Should GenAI tools show a warning on every answer? Suggest a design that helps users trust but verify.

UNIT REVIEW: AI DISCOVER

Name: _____ Date: _____

Circle the right answer!

Q1. Inputs AI uses are...

a) Features | b) Balloons | c) Biscuits

Q2. LLMs predict the next...

a) Weather | b) Token | c) Dice

Q3. Memorising, not learning, is...

a) Accuracy | b) Running | c) Overfitting

Q4. A confident wrong answer is a...

a) Hallucination | b) Prize | c) Fact

Q5. How much a feature matters is its...

a) Tune | b) Weight | c) Colour

Q6. Image AI starts from...

a) Pizza | b) Paint | c) Noise

WORD SEARCH: AI DISCOVER

Find these words. They go across → or down ↓.

FEATURE

WEIGHT

TOKEN

LLM

MODEL

PIXEL

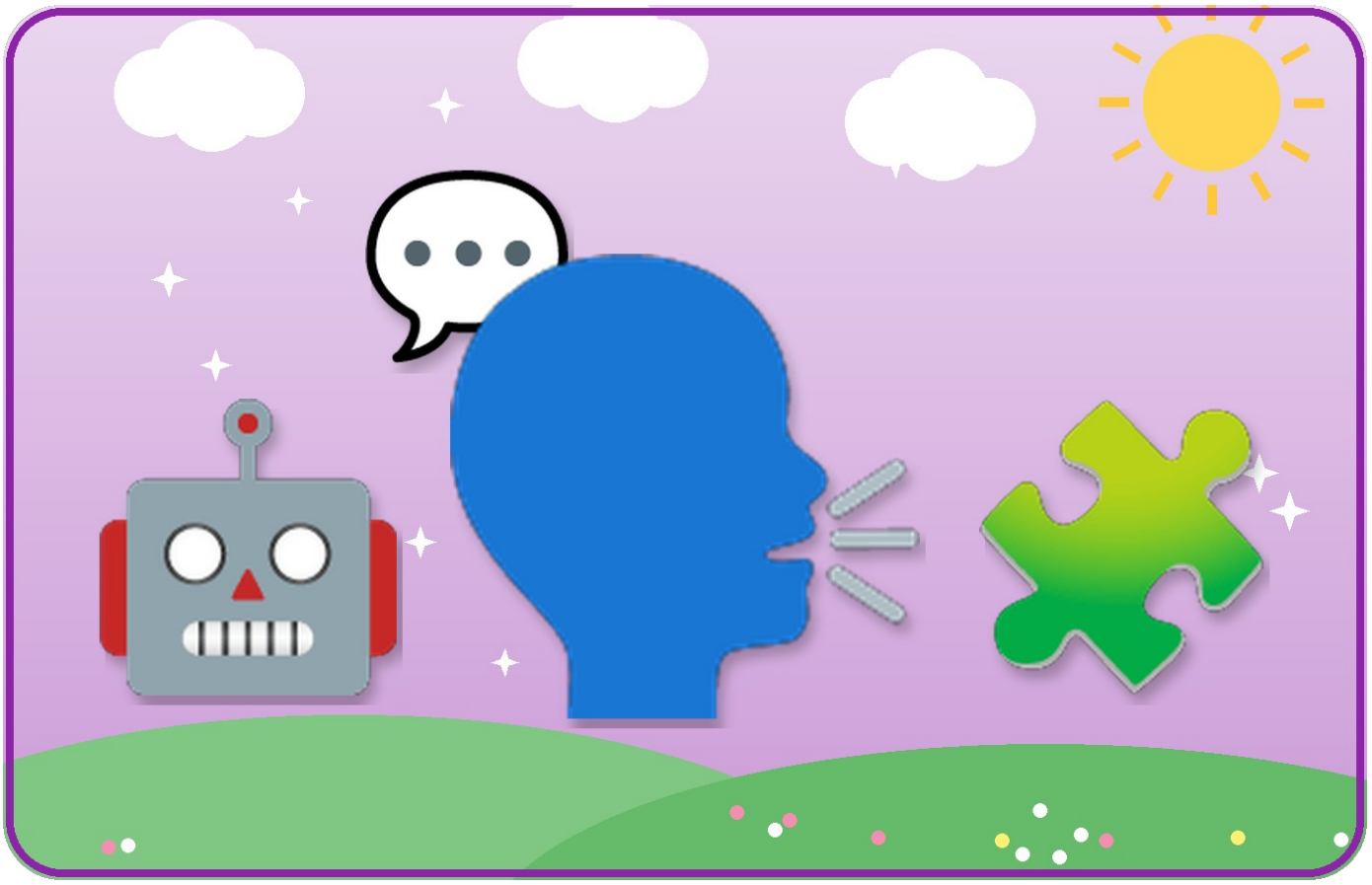
A	A	T	A	N	G	O	A	S	H
P	T	F	E	A	T	U	R	E	R
T	O	H	M	H	H	P	K	A	O
T	K	D	F	P	I	X	E	L	K
D	E	L	S	O	S	G	K	K	U
L	N	R	S	N	U	B	R	H	N
L	O	F	M	O	D	E	L	M	T
M	M	C	W	E	I	G	H	T	P
S	D	F	S	N	M	R	A	R	B
K	W	U	U	N	F	F	S	H	A

I finished AI DISCOVER! Teacher's sign: _____

MONTH 2

AI CONNECT

Think • Instruct • Code



Area: Communicate with AI

In this unit, you will explore:

Topic 3

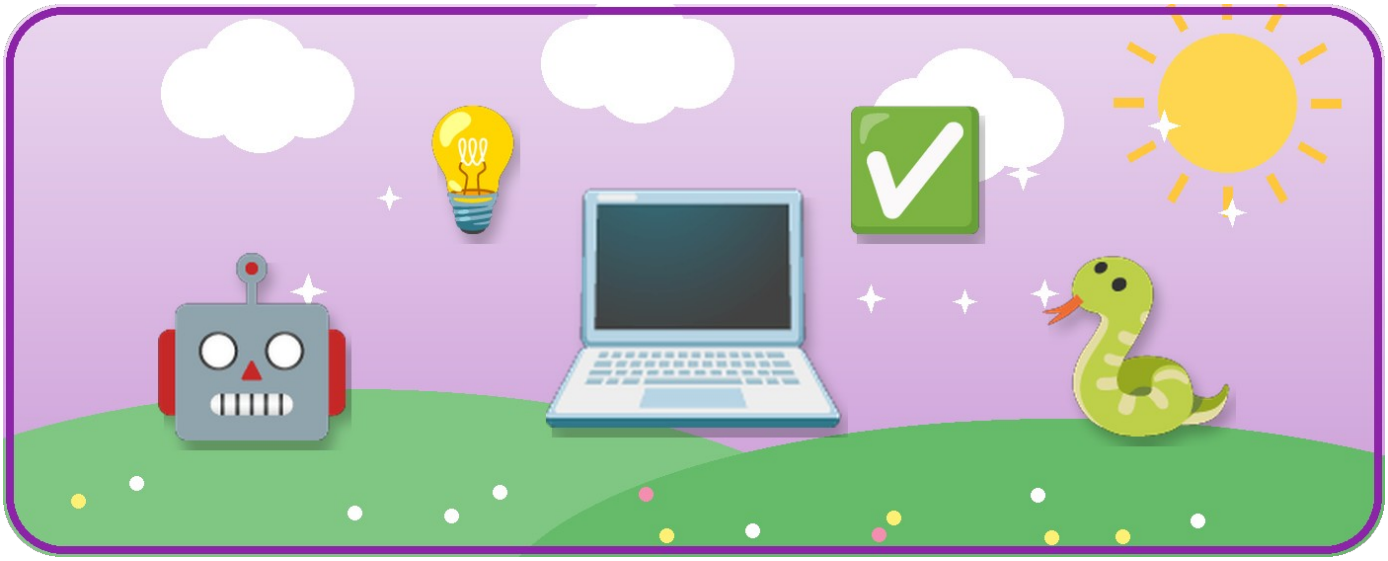
AI-Assisted Coding

Topic 4

Code → Test → Improve

TOPIC 3 • MONTH 2 • COMMUNICATE WITH AI

AI-Assisted Coding



LET'S LEARN

- AI code assistants suggest and explain code.
- You still must read, understand and test every line.
- AI is a pair programmer, not a replacement!

Smart ways to use AI

Explain code

Find bugs

Suggest ideas

Add comments

MAGIC WORDS

Pair programming Coding together

Comment Notes in code

Code review Checking code

💬 TECH TALK: AI-Assisted Coding

💬 **GAIO Squad** Riya, Kabir, Bolt • online

👦 **Kabir**
Bolt, an AI assistant wrote my whole function in seconds ⚡!
10:02

🤖 **Bolt**
Nice! But did you read and understand every line 👀?
10:04 ✓✓

👦 **Kabir**
Umm... not really.
10:06

🤖 **Bolt**
Then you can't fix it when it breaks. Treat AI as a pair programmer 👥: it suggests, you decide.
10:08 ✓✓

👧 **Riya**
How do we use it well?
10:10

🤖 **Bolt**
Ask it to explain code, add comments 💬 and suggest tests. Then run them yourself 🧪.
10:12 ✓✓

👦 **Kabir**

Can AI code be wrong?

10:14

🤖 **Bolt**

Yes: bugs, outdated methods, even security holes 🔓. Do a code review before using it.

10:16 ✓✓

👧 **Riya**

Read, understand, test, own it ✅!

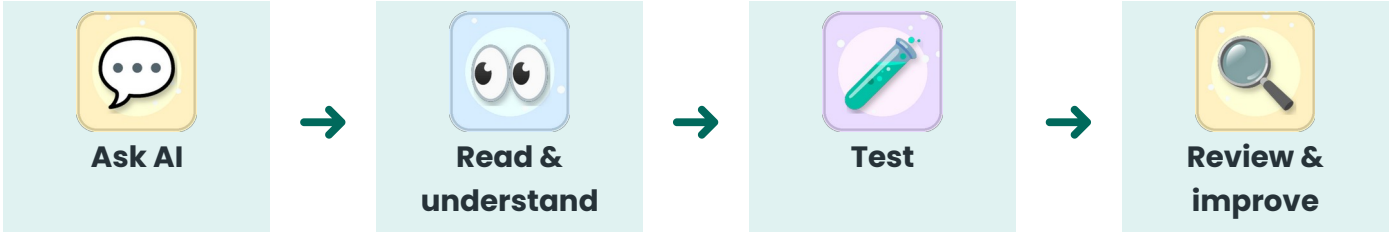
10:18

😊 Type a message...

📌 **Pinned by Bolt**

AI code assistants suggest and explain code, but you must read, understand, test and review every line. AI is a pair programmer, not a replacement.

🔄 How it works



INDIA SPOTLIGHT

India has one of the world's largest developer communities, and many developers now use AI assistants in daily coding.

AI CHRONICLES: The Bug Behind the Shortcut

Riya | Kabir | Code Assistant

SCENE 1

Riya used an AI assistant to write a login page for the school club website in minutes.

Kabir reviewed it line by line. "Wait, passwords are stored as plain text 🔓 !"

SCENE 2

SCENE 3

Anyone who saw the database could read every password. A serious security hole.

Riya asked the assistant to explain secure storage. It suggested hashing; she read the docs to confirm.

SCENE 4

SCENE 5

They rewrote it, added tests and asked their teacher for a final code review.

The site launched safely. "AI is my pair programmer, not my replacement," Riya said.

SCENE 6

🔮 What happens next? Tick ✓ one!

☐	☐	☐
They add two-factor login	**Riya stops reviewing AI code**	**The passwords hide themselves**

🧐 Story check: answer the questions.

1. What problem did Kabir find?

2. Why is it dangerous?

3. What does 'pair programmer' mean here?

Rate this story: shade the stars.

☆ ☆ ☆ ☆ ☆

LOOK AROUND YOU: AI coding helpers can...

Explain error messages

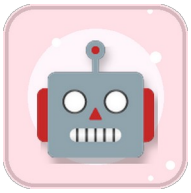
Convert a loop to a function

Write comments

Suggest test cases

Translate code languages

Clean messy code



BOLT SAYS:
Never copy code you can't explain. Ask me "Why?" until you understand!

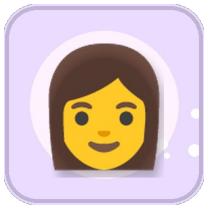
FUN FACT!
"Pair programming" means two people code together. Now one partner can be an AI!

👀 **I spy AI! Draw one AI helper you saw today.**

LET'S DO IT! Explain-the-Code Challenge

You need:

Code snippet



Teacher's AI tool

Pairs

Steps:

1. Read a 6-line Python snippet. Guess what it does.
2. Ask the AI tool to explain it (with teacher).
3. Compare: was your guess right?
4. Add your own comments to each line.

THINK & TALK

Why must you understand code before using it?

Draw what you did!

WORKSHEET: AI-Assisted Coding (Part A)

Name: _____ Date: _____

1. Good or bad use of AI coding help? Tick ✓.

	Good	Bad
Ask AI to explain an error		
Submit AI code you can't explain		
Ask AI for test ideas		
Skip testing AI code		

2. What does this code print? Use the word box.

WORD BOX: Hi Meera ✿ 15 ✿ Pass

name = "Meera"; print(f"Hi {name}") → ____

print(sum([4, 5, 6])) → ____

m = 70; print("Pass" if m >= 35 else "Fail") → ____

WORKSHEET: AI-Assisted Coding (Part B)

Name: _____ Date: _____

1. Match the request to how AI helps.

Explain this error	●	●	Notes in code
Suggest tests	●	●	Neater code
Add comments	●	●	Test ideas
Clean my code	●	●	Simple explanation

2. Write a 3-line Python program and comment each line!

My program: _____

PUZZLE FUN: AI-Assisted Coding

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.

A				
B				
C				

2. Define it! Write each term in your own words with an example.

Pair programming

Meaning: _____

Example: _____

Comment

Meaning: _____

Example: _____

Code review

Meaning: _____

Example: _____

🧪 DEMO LAB: Online Python editor + AI assistant (with teacher)

Name: _____ Date: _____

🎯 **Goal:** Pair-program with AI, then review its code.

🔗 **Open:** Google Colab or any editor with AI help | 👩‍🏫 **With a teacher or parent**

👣 Steps

1		Ask AI: 'Write a Python function to check if a word is a palindrome.'
2		Read every line. Add a comment above each in your own words.
3		Test with 'madam', 'Madam', 'race car' and ''.
4		Find at least one weakness (case, spaces, empty input).
5		Fix it yourself, then ask AI to review your fix.

🔬 Record your observations

Weakness found:	
Your fix:	
AI review comment:	

💭 **Reflect:** Why must you understand AI-written code before using it?

? QUIZ TIME: AI-Assisted Coding

Circle the right answer!

Q1. AI coding assistants are like a...

a) Pair programmer

b) Broom

c) Toy

Q2. Before using AI code, you must...

a) Copy it

b) Understand & test it

c) Hide it

Q3. Notes inside code are...

a) Songs

b) Pizza

c) Comments

Q4. f"Hi {name}" is an...

a) f-string

b) Fish

c) Error

Q5. Checking someone's code is...

a) Singing

b) Code review

c) Napping

MY SCORE: ____ / 5. Shade one star per correct answer.

☆ ☆ ☆ ☆ ☆

✅ TRUE OR FALSE: AI-Assisted Coding

Look at the picture. Tick ✓ TRUE or FALSE.

		👍 TRUE	👎 FALSE
	AI can be a pair programmer.		
	Submitting AI code you can't explain is fine.		
	Comments are notes inside code.		
	AI can suggest test cases.		
	AI code never needs testing.		
	f"Hi {name}" is an f-string.		

HOME CONNECT

Explain one piece of code (or a recipe!) line by line to a family member.

🔢 Self-assessment: tick one.

I know it well!

I know some.

I need practice.

EXERCISE: AI-Assisted Coding

Name: _____ Date: _____

A. Tick (✓) the correct answer.

1. AI coding help where you still decide is ____ programming.

(a) pair ☐ **(b)** pirate ☐ **(c)** paper ☐

2. Notes inside code explaining it are ____.

(a) errors ☐ **(b)** comments ☐ **(c)** commands ☐

3. Checking code quality before use is a code ____.

(a) race ☐ **(b)** rewrite ☐ **(c)** review ☐

4. Storing passwords as plain text is a ____ risk.

(a) security ☐ **(b)** colour ☐ **(c)** speed ☐

B. Fill in the blanks. Use the word box.

understand **explain** **bugs** **hashed**

1. Read and ____ every line of AI code.

2. AI can ____ code it writes.

3. AI code may have ____.

4. Passwords should be ____, not plain text.

C. Write T for True and F for False.

1. AI assistants can suggest code. [ ]

2. AI-written code never needs testing. [ ]

3. Comments help others understand code. [ ]

4. You can't fix code you don't understand. [ ]

D. Match the following. Write the letter.

Column A		Column B
1. Pair programming	____	**(a)** Checks quality
2. Comment	____	**(b)** AI suggests, you decide
3. Code review	____	**(c)** Weakness attackers use
4. Security hole	____	**(d)** Explains code

E. Answer in one word.

1. What explains code in plain words? _____

2. What checks code before use? _____

3. AI as coding partner is ____ programming. _____

F. Answer these questions.

1. List four ways to use AI code assistants responsibly.

2. Why is understanding code more important than writing it fast?

G. Case-based questions. Read and answer.

An AI assistant wrote a login page that stored passwords as plain text. Kabir's review caught it; they rewrote it, added tests and got a teacher review.

1. Plain-text passwords are a ____.

(a) security risk ☐ **(b)** feature ☐ **(c)** comment ☐

2. The problem was found through ____.

(a) luck ☐ **(b)** code review ☐ **(c)** a virus ☐

3. This shows AI is a ____.

(a) replacement ☐ **(b)** boss ☐ **(c)** pair programmer ☐

H. THINK DEEPER

Write a checklist of five things to verify before using any AI-generated code in a real app.

TOPIC 4 • MONTH 2 • COMMUNICATE WITH AI

Code → Test → Improve

LET'S LEARN

⭐ Great code is written, tested and improved again and again.

⭐ Tests check normal inputs AND tricky edge cases.

⭐ try/except handles errors without crashing!

👀 The improve loop

Code

Test

Fix

Improve

MAGIC WORDS

Unit test Check one small part

Edge case Unusual input

Exception Error during running

💬 TECH TALK: Code → Test → Improve

💬 GAIO Squad Riya, Kabir, Bolt • online

👧 Riya

My grade calculator works for 85. Done 😎?

10:02

🤖 Bolt

Not yet! Write unit tests 🧪: small checks that each function gives the right answer.

10:04 ✓✓

👦 Kabir

Which inputs?

10:06

🤖 Bolt

Normal ones, plus edge cases ⚠️: 0, 100, 90 exactly, negative numbers, even text.

10:08 ✓✓

👧 Riya

If someone types 'abc', it crashes 💥!

10:10

🤖 Bolt

Use try/except 🛡️: try the conversion; except ValueError, print 'Please enter a number'.

10:12 ✓✓

Kabir
So exceptions are handled, not ignored.
10:14

Bolt
Right. Then refactor 🔧: make code cleaner without changing what it does.
10:16 ✓✓

Riya
Code, test, improve, repeat 🔁!
10:18

😊 Type a message...

📌 **Pinned by Bolt**

Great code is tested with normal and edge cases. try/except handles exceptions without crashing. Refactoring improves code without changing behaviour.

🔄 How it works

INDIA SPOTLIGHT

Software companies in India's tech hubs run thousands of automated tests every day before releasing apps.

AI CHRONICLES: The App That Wouldn't Crash

Kabir

Riya

Hackathon Judge

SCENE 1

At a state hackathon, Kabir's team built a fee-calculator app for their school 🧾.

A judge tried to break it: she typed 'abc', -500 and left the field empty 💥.

SCENE 2

SCENE 3

The app crashed on 'abc'. Panic! They had only tested normal numbers.

In 20 minutes they added try/except and unit tests for every edge case.

SCENE 4

SCENE 5

They refactored messy code into clean functions without changing results.

The judge retested: no crash. "Testing is what makes code professional," she said. They won second place!

SCENE 6

🔮 What happens next? Tick ✓ one!

☐	☐	☐
They write tests first next time	**They ignore edge cases again**	**The app learns to dodge judges**

🧐 Story check: answer the questions.

1. What inputs crashed the app?

2. How did they fix it?

3. What is refactoring?

Rate this story: shade the stars.

☆ ☆ ☆ ☆ ☆

LOOK AROUND YOU: Why testing matters

Banking apps must be exact

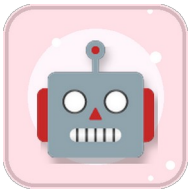
Rocket code can't fail

Medical devices need safety

Games need no crashes

Vote counts must be right

Traffic systems must be reliable



BOLT SAYS:
assert add(2, 3) == 5 — if it fails, I shout! That's a unit test.

FUN FACT!
Big software companies run thousands of automatic tests every time code changes!

I spy AI! Draw one AI helper you saw today.

LET'S DO IT! Break My Program!

You need:

Simple program

Test cards

Teams

Steps:

1		Team A writes: divide(a, b) returns a / b.
2		Team B tries tricky inputs: b = 0, text, negatives.
3		Team A adds try/except to handle errors.
4		Repeat until it can't be broken!

THINK & TALK

Which edge case broke the program first?

Draw what you did!

WORKSHEET: Code → Test → Improve (Part A)

Name: _____ Date: _____

1. The improve loop. Write 1, 2, 3, 4.

Improve ☐

Write code ☐

Fix bugs ☐

Test ☐

2. Edge case for a marks program (0–100)? Tick ✓.

	Edge case	Normal
0 marks = 0		
5 marks = 55		
💯 marks = 100		
➖ marks = -5		
7 marks = 72		

WORKSHEET: Code → Test → Improve (Part B)

Name: _____ Date: _____

1. Complete the safe code. Use the word box.

WORD BOX: try ✿ except ✿ assert

____: result = a / b

____ ZeroDivisionError: print("No!")

____ divide(6, 2) == 3

2. Match the term to its meaning.

Unit test ●

● **Improve again & again**

Edge case ●

● **Unusual input**

try/except ●

● **Handle errors**

Iterate ●

● **Check one part**

PUZZLE FUN: Code → Test → Improve

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.

A				
B				
C				

2. Define it! Write each term in your own words with an example.

Unit test

Meaning: _____

Example: _____

Edge case

Meaning: _____

Example: _____

Exception

Meaning: _____

Example: _____

🧪 DEMO LAB: Online Python editor

Name: _____ Date: _____

🎯 **Goal:** Write unit tests and handle exceptions with try/except.

🔗 **Open:** any browser Python editor / Google Colab | 👩‍🏫 **With a teacher or parent**

👣 Steps

1		def grade(m): return 'A' if m >= 90 else 'B' if m >= 75 else 'C'
2		assert grade(95) == 'A'; assert grade(90) == 'A'; assert grade(74) == 'C'
3		Add edge tests: grade(0), grade(100), grade(-5). What should -5 do?
4		Wrap input: try: m = int(input()) except ValueError: print('Enter a number')
5		Refactor so invalid marks (< 0 or > 100) return 'Invalid'.

🔬 Record your observations

Tests passed first time:	
Edge case that failed:	
Your refactored rule:	

💭 **Reflect:** Why do professional teams write tests before releasing apps?

? QUIZ TIME: Code → Test → Improve

Circle the right answer!

Q1. Checking one small part is a...

a) Unit test

b) Party

c) Cake

Q2. An unusual input is an...

a) Average

b) Edge case

c) Echo

Q3. try/except helps code...

a) Crash

b) Sleep

c) Handle errors

Q4. divide(5, 0) is an...

a) Edge case

b) Normal case

c) Party

Q5. Great code is improved...

a) Once

b) Again & again

c) Never

MY SCORE: ____ / 5. Shade one star per correct answer.

☆ ☆ ☆ ☆ ☆

✅ TRUE OR FALSE: Code → Test → Improve

Look at the picture. Tick ✓ TRUE or FALSE.

		👍 TRUE	👎 FALSE
	A unit test checks one small part.		
	divide by zero is an edge case.		
	try/except makes programs crash more.		
	Great code improves over time.		
	Testing once is always enough.		
	Rocket software needs careful testing.		

HOME CONNECT

Test a family gadget's "edge cases" (e.g., what happens if the remote has no batteries?)!

🔢 Self-assessment: tick one.

I know it well!

I know some.

I need practice.

EXERCISE: Code → Test → Improve

Name: _____ Date: _____

A. Tick (✓) the correct answer.

1. A small check that one function works is a ____ test.

(a) unit ☐ **(b)** union ☐ **(c)** uniform ☐

2. Extreme or unusual inputs are ____ cases.

(a) easy ☐ **(b)** edge ☐ **(c)** empty ☐

3. Python handles errors without crashing using ____.

(a) if/else ☐ **(b)** for/in ☐ **(c)** try/except ☐

4. Cleaning code without changing behaviour is ____.

(a) refactoring ☐ **(b)** rewinding ☐ **(c)** removing ☐

B. Fill in the blanks. Use the word box.

Value condition negative repeat

1. int('abc') raises a ____Error.

2. assert checks a ____.

3. Test 0, 100 and ____ numbers.

4. Code, test, improve, ____.

C. Write T for True and F for False.

1. Edge cases often reveal bugs. [ ]

2. Testing one normal input is enough. [ ]

3. try/except can prevent crashes. [ ]

4. Refactoring changes what the program outputs. [ ]

D. Match the following. Write the letter.

Column A		Column B
1. Unit test	____	**(a)** Runtime error
2. Edge case	____	**(b)** Checks one function
3. Exception	____	**(c)** Clean up code
4. Refactor	____	**(d)** Extreme input

E. Answer in one word.

1. Which keyword pair handles errors? _____

2. What checks one function? _____

3. What error does int('abc') raise? _____

F. Answer these questions.

1. Write three unit tests for a function that checks if a number is even.

2. Why should apps handle exceptions?

G. Case-based questions. Read and answer.

A judge typed 'abc', -500 and an empty field into a fee calculator. It crashed on 'abc'. The team added try/except, unit tests and refactored the code.

1. 'abc' and -500 are ____.

(a) edge cases ☐ **(b)** comments ☐ **(c)** features ☐

2. The crash was prevented using ____.

(a) print ☐ **(b)** try/except ☐ **(c)** while ☐

3. Cleaning the code was ____.

(a) deleting ☐ **(b)** copying ☐ **(c)** refactoring ☐

H. THINK DEEPER

Pick an app you use. List five edge cases its developers must have tested and what should happen for each.

🏅 UNIT REVIEW: AI CONNECT

Name: _____ Date: _____

Circle the right answer!

Q1. An unusual input is an...

a) Edge case

b) Average

c) Echo

Q2. Notes inside code are...

a) Songs

b) Comments

c) Pizza

Q3. try/except helps code...

a) Crash

b) Sleep

c) Handle errors

Q4. AI coding helpers are like...

a) Pair programmers

b) Brooms

c) Toys

Q5. Checking one small part is a...

a) Party

b) Unit test

c) Cake

Q6. Before using AI code, you...

a) Copy it

b) Hide it

c) Understand & test

WORD SEARCH: AI CONNECT

Find these words. They go across → or down ↓.

COMMENT

CODE REVIEW

UNIT TEST

EDGE CASE

EXCEPTION

PYTHON

P	C	O	S	M	U	E	M	M	P
F	O	N	C	P	S	X	H	R	I
R	D	S	O	S	M	C	E	U	P
P	E	M	M	U	T	E	D	N	P
R	R	P	M	H	L	P	G	I	F
W	E	Y	E	I	R	T	E	T	K
K	V	T	N	S	T	I	C	T	S
S	I	H	T	W	U	O	A	E	O
K	E	O	G	R	S	N	S	S	M
W	W	N	C	L	A	G	E	T	D

I finished AI CONNECT! Teacher's sign: _____

MONTH 3

AI SOLVE

Problems • Solutions • Impact

Area: AI Applications

In this unit, you will explore:

Topic 5

AI at Work

Topic 6

AI Solving Real Problems

TOPIC 5 • MONTH 3 • AI APPLICATIONS

AI at Work

LET'S LEARN

- AI is changing how almost every industry works.
- It automates repetitive tasks and assists experts.
- Humans focus on creativity, care and judgement!

AI at work

Automates tasks

Assists experts

Analyses data

Speeds work

MAGIC WORDS

Automation
Machines doing tasks

Augmentation
AI helping humans

Industry A field of business

💬 TECH TALK: AI at Work

💬 GAIO Squad Riya, Kabir, Bolt • online

👦 **Kabir**

Bolt, my uncle works in a bank. Will AI replace him 😟?

10:02

🤖 **Bolt**

AI automates repetitive tasks 🤖, like checking forms. But it augments people too.

10:04 ✓✓

👧 **Riya**

Augments?

10:06

🤖 **Bolt**

Makes them stronger 💪. AI flags risky loans; your uncle decides and talks to customers.

10:08 ✓✓

👦 **Kabir**

Which industries use AI?

10:10

🤖 **Bolt**

Healthcare 🏥, farming 🌾, factories 🏭, retail, logistics 🚚, media and more.

10:12 ✓✓

👧 **Riya**

What stays human?

10:14

🤖 **Bolt**

Creativity 🎨, care ❤️, judgement and leadership. Those are hard to automate.

10:16 ✓✓

👦 **Kabir**

So workers must learn to work WITH AI 🤝.

10:18

😊 Type a message...

📌 **Pinned by Bolt**

AI automates repetitive tasks and augments experts across industries. Humans focus on creativity, care, judgement and leadership.

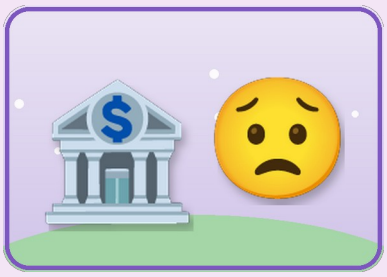
🔄 How it works

INDIA SPOTLIGHT

Indian manufacturers, banks and hospitals increasingly use AI for quality checks, fraud detection and diagnostics.

AI CHRONICLES: Uncle Ravi's New Partner

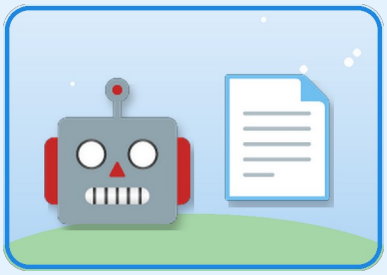
Riya | Uncle Ravi | Bank AI



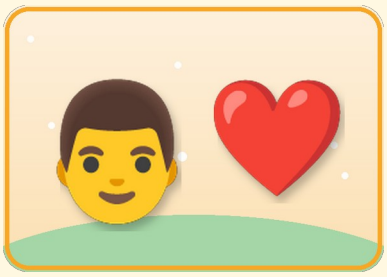
SCENE 1

Uncle Ravi, a bank loan officer, worried when the bank introduced an AI system.

The AI checked thousands of forms in minutes, a task that used to take him days.



SCENE 2



SCENE 3

With time saved, he met farmers personally, explained loans and understood their needs.

Once, AI rejected a farmer because of a missing record. Ravi investigated and approved it fairly.

SCENE 4

SCENE 5

He took a short course on reading AI reports and became the branch's AI champion 🏅.

Riya wrote in her essay: "AI took the paperwork. Uncle kept the people."

SCENE 6

🔮 What happens next? Tick ✓ one!

☐	☐	☐
Uncle Ravi trains other officers	**Uncle Ravi quits in fear**	**The AI applies for a loan**

🧐 Story check: answer the questions.

1. What task did AI automate?

2. How did AI augment Uncle Ravi's work?

3. Why did the farmer's case need human judgement?

Rate this story: shade the stars.

☆ ☆ ☆ ☆ ☆

LOOK AROUND YOU: AI in Indian industries

Factories: quality checks

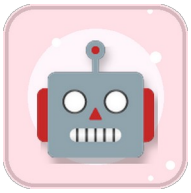
Banks: fraud detection

Retail: stock prediction

Logistics: route planning

Hospitals: scan reading

Call centres: chatbots



BOLT SAYS:

I'm great at repeating tasks. You're great at new ideas and caring for people!

FUN FACT!

AI cameras in factories can spot tiny defects faster than the human eye!

I spy AI! Draw one AI helper you saw today.

LET'S DO IT! Job Task Sort

You need:

Task cards

Sort chart

Groups

Steps:

1

Read task cards: sort files, comfort patient, design logo...

2

Sort: AI can do / AI can help / humans only.

3

Debate the tricky ones.

4

Which human skills seem safest?

THINK & TALK

Which job tasks need a human heart?

Draw what you did!

WORKSHEET: AI at Work (Part A)

Name: _____ Date: _____

1. Match the industry to its AI use.

Bank

Factory

Delivery

Shop

Best routes

Stock prediction

Fraud detection

Defect spotting

2. Automation or human judgement? Tick ✓.

	Automate	Human
Sorting thousands of files		
Comforting a patient		
Checking bill totals		
A judge's final decision		

WORKSHEET: AI at Work (Part B)

Name: _____ Date: _____

1. Circle the tasks AI can automate well.

Sorting files

Comforting a child

Checking totals

Original art direction

Filtering spam

Spotting defects

2. Fill in the blanks. Use the word box.

WORD BOX: automation ✿ augmentation ✿ industry

Machines doing tasks is _____.

AI helping humans is _____.

A field of business is an _____.

PUZZLE FUN: AI at Work

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.

A				
B				
C				

2. Define it! Write each term in your own words with an example.

Automation

Meaning: _____

Example: _____

Augmentation

Meaning: _____

Example: _____

Industry

Meaning: _____

Example: _____

🧪 DEMO LAB: Industry interview + mapping (unplugged)

Name: _____ Date: _____

🎯 **Goal:** Discover how AI is changing a real job.

🔗 **Open:** interview a family member / neighbour | 👩‍🏫 **With a teacher or parent**

👣 Steps

1. Interview someone working (any field). Ask about their daily tasks.
2. List tasks that are repetitive.
3. Mark tasks AI could automate and tasks AI could augment.
4. Mark tasks that need creativity, care or judgement.
5. Draw a 3-column chart: Automate / Augment / Human.

🔬 Record your observations

Job studied:	
Task AI could automate:	
Task that stays human:	

💭 **Reflect:** What new skills should this worker learn to work with AI?

? QUIZ TIME: AI at Work

Circle the right answer!

Q1. Machines doing tasks is...

a) Automation

b) Art

c) Rest

Q2. AI helping humans is...

a) Destruction

b) Augmentation

c) Hiding

Q3. Banks use AI for...

a) Music

b) Pizza

c) Fraud detection

Q4. Humans focus on...

a) Creativity & care

b) Only repetition

c) Sleeping

Q5. Factories use AI cameras to spot...

a) Rainbows

b) Defects

c) Birds

MY SCORE: ____ / 5. Shade one star per correct answer.

☆ ☆ ☆ ☆ ☆

✅ TRUE OR FALSE: AI at Work

Look at the picture. Tick ✓ TRUE or FALSE.

		👍 TRUE	👎 FALSE
🔁	AI can automate repetitive tasks.		
🤝	Augmentation means AI helping humans.		
🤗	AI is better at caring than humans.		
🏦	Banks use AI to detect fraud.		
🎓	Workers can learn new AI skills.		
🤐	AI is used in no Indian industries.		

HOME CONNECT

Ask a working family member: which tasks in their job could AI help with?

🔢 Self-assessment: tick one.

😀	🙂	😐
I know it well!	**I know some.**	**I need practice.**

EXERCISE: AI at Work

Name: _____ Date: _____

A. Tick (✓) the correct answer.

1. AI doing repetitive tasks by itself is ____.

(a) automation ☐ **(b)** augmentation ☐ **(c)** animation ☐

2. AI making experts more capable is ____.

(a) automation ☐ **(b)** augmentation ☐ **(c)** aggravation ☐

3. Tasks needing care and judgement are best done by ____.

(a) printers ☐ **(b)** robots only ☐ **(c)** humans ☐

4. AI is used in healthcare, farming and ____.

(a) logistics ☐ **(b)** nowhere else ☐ **(c)** only games ☐

B. Fill in the blanks. Use the word box.

decide automate with human

1. AI flags risks; humans ____.

2. Creativity and care are hard to ____.

3. Workers should learn to work ____ AI.

4. AI frees time for ____ work.

C. Write T for True and F for False.

1. AI can automate form checking. [ ]

2. AI will make every human skill useless. [ ]

3. Augmentation helps experts work better. [ ]

4. Only IT companies use AI. [ ]

D. Match the following. Write the letter.

Column A		Column B
1. Automation	____	**(a)** Human decision
2. Augmentation	____	**(b)** AI does task alone
3. Judgement	____	**(c)** Sector of work
4. Industry	____	**(d)** AI assists human

E. Answer in one word.

1. AI doing tasks alone is? _____

2. AI assisting experts is? _____

3. Name one human strength. _____

F. Answer these questions.

1. Differentiate automation and augmentation with examples.

2. Which skills will stay valuable as AI spreads?

G. Case-based questions. Read and answer.

A bank's AI checks loan forms in minutes. Uncle Ravi now meets farmers personally, and once overruled an unfair AI rejection after investigating.

1. Checking forms is _____.

(a) automation ☐ **(b)** augmentation ☐ **(c)** imagination ☐

2. Meeting farmers shows _____ skills.

(a) machine ☐ **(b)** human ☐ **(c)** printing ☐

3. Overruling the AI shows _____.

(a) obedience ☐ **(b)** speed ☐ **(c)** human judgement ☐

H. THINK DEEPER

Choose one industry. Predict how AI will change it by 2035 and which new jobs may appear.

TOPIC 6 • MONTH 3 • AI APPLICATIONS

AI Solving Real Problems

LET'S LEARN

- Design thinking solves problems by focusing on people.
- Empathise, define, ideate, prototype, test.
- AI is a tool. The problem and the people come first!

Design thinking

Empathise

Define

Ideate

Prototype & test

MAGIC WORDS

Empathy Feeling with others

Prototype First test model

Ideate Generate ideas

💬 TECH TALK: AI Solving Real Problems

💬 GAIO Squad Riya, Kabir, Bolt • online

👧 Riya
Let's build an AI app for elderly people 👵!
10:02

🤖 Bolt
Wait! First empathise ❤️: talk to them. What do THEY struggle with?
10:04 ✓✓

👦 Kabir
Grandma forgets her medicines 💊.
10:06

🤖 Bolt
Now define the problem clearly: 'Seniors living alone miss medicine doses.'
10:08 ✓✓

👧 Riya
Then ideate 💡: alarms, voice reminders, pill-box sensors...
10:10

🤖 Bolt
Pick one and build a quick prototype 🛠️, even on paper. Then test with real users 🧪.
10:12 ✓✓

👦 **Kabir**

Where does AI fit?

10:14

🤖 **Bolt**

Maybe a voice assistant in her language. But the person and the problem come first 👥.

10:16 ✓✓

👧 **Riya**

Empathise, define, ideate, prototype, test 🔁!

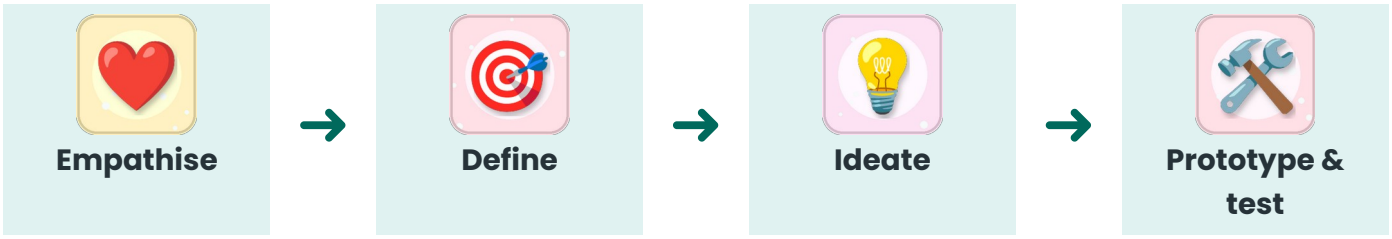
10:18

😊 Type a message...

📌 **Pinned by Bolt**

Design thinking puts people first: empathise, define, ideate, prototype and test. AI is a tool to solve the real problem.

🔄 How it works



INDIA SPOTLIGHT

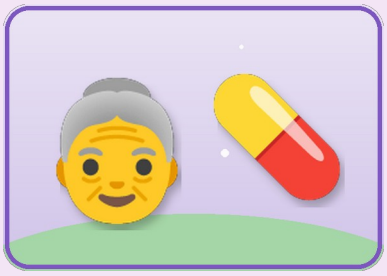
Atal Innovation Mission's tinkering labs and challenges encourage Indian students to solve community problems with design thinking.

AI CHRONICLES: Paati's Pill Pal

Kabir

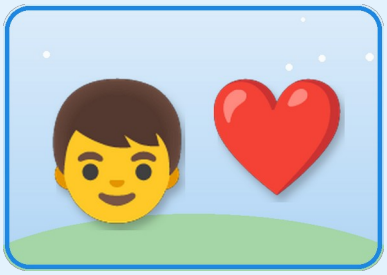
Paati

Riya


SCENE 1

Kabir's Paati lived alone and often missed her medicines. The family worried 😟.

Instead of building an app at once, Kabir spent a day with her: empathise first!


SCENE 2

SCENE 3

Insight: she couldn't read small text but loved her radio 📻. Problem defined.

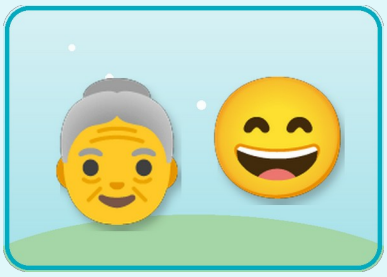
They prototyped a voice reminder in Tamil that played from a cheap speaker, plus a colour-coded pill box.

SCENE 4

SCENE 5

Testing showed the voice was too fast. They slowed it and added her favourite song as the alert 🎵.

Paati hasn't missed a dose in a month. "People first, then technology," Riya noted.



SCENE 6

🔮 What happens next? Tick ✓ one!

☐	☐	☐
They make it for 10 more elders	**They build an app Paati can't read**	**The pill box starts singing alone**

🧐 Story check: answer the questions.

1. What did Kabir do before building?

2. What insight changed their design?

3. What did testing reveal?

Rate this story: shade the stars.

☆ ☆ ☆ ☆ ☆

LOOK AROUND YOU: Real problems, AI solutions

Water leaks → sensor alerts

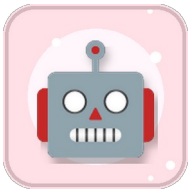
Food waste → demand forecast

Late buses → live tracking

Lonely elders → check-in calls

Weak reading → reading tutor

Sick cattle → photo diagnosis



BOLT SAYS:

First ask: who has the problem? Talk to them before building anything!

FUN FACT!

Students across India build AI solutions in national innovation challenges and hackathons!

I spy AI! Draw one AI helper you saw today.

LET'S DO IT! Mini Hackathon

You need:

Problem cards

Poster paper

45-minute timer

Steps:

1		Interview a classmate about a school problem.
2		Define the problem in one sentence.
3		Brainstorm 10 ideas. Pick 1 that uses AI wisely.
4		Sketch a paper prototype and test it on 3 people.

THINK & TALK

Did testing change your idea? How?

Draw what you did!

WORKSHEET: AI Solving Real Problems (Part A)

Name: _____ Date: _____

1. Design thinking steps. Write 1, 2, 3, 4.

Prototype & test ☐

Empathise ☐

Ideate ☐

Define ☐

2. Match the problem to an AI solution.

Water leaks ●

●

Photo diagnosis

Food waste ●

●

Reading tutor

Sick cattle ●

●

Sensor alerts

Weak reading ●

●

Demand forecast

WORKSHEET: AI Solving Real Problems (Part B)

Name: _____ Date: _____

1. Match the design thinking step to the action.

Empathise ●

● Brainstorm 10 ideas

Define ●

● Interview users

Ideate ●

● Build a quick model

Prototype ●

● Write one problem sentence

2. Sketch a prototype that solves a problem in your school!

Problem: _____

PUZZLE FUN: AI Solving Real Problems

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.

A				
B				
C				

2. Define it! Write each term in your own words with an example.

Empathy

Meaning: _____

Example: _____

Prototype

Meaning: _____

Example: _____

Ideate

Meaning: _____

Example: _____

🧪 DEMO LAB: Design thinking sprint (unplugged)

Name: _____ Date: _____

🎯 **Goal:** Solve a real school problem with design thinking.

🔗 **Open:** sticky notes + chart paper | 👩‍🏫 **With a teacher or parent**

👣 Steps

1. Interview 3 classmates about one daily problem (e.g., heavy bags).
2. Write a problem statement: '[User] needs [need] because [insight].'
3. Ideate: 10 ideas in 5 minutes. No judging.
4. Build a paper or cardboard prototype of the best idea.
5. Test with 2 users and note feedback.

🔬 Record your observations

Problem statement:	
Chosen idea:	
User feedback:	

💭 **Reflect:** Where (if at all) could AI improve your solution?

? QUIZ TIME: AI Solving Real Problems

Circle the right answer!

Q1. Design thinking starts with...

a) Empathy

b) Code

c) Money

Q2. A first test model is a...

a) Cake

b) Prototype

c) Trophy

Q3. Generating ideas is to...

a) Sleep

b) Delete

c) Ideate

Q4. What comes first?

a) People & problem

b) AI tool

c) Logo

Q5. Students build solutions in...

a) Video games only

b) Hackathons

c) Naps

MY SCORE: ____ / 5. Shade one star per correct answer.

☆ ☆ ☆ ☆ ☆

✅ TRUE OR FALSE: AI Solving Real Problems

Look at the picture. Tick ✓ TRUE or FALSE.

		👍 TRUE	👎 FALSE
	Design thinking starts with empathy.		
	Pick the AI tool before knowing the problem.		
	A prototype is a first test model.		
	Ideate means generate ideas.		
	Students can join innovation challenges.		
	Testing ideas with users is a waste of time.		

HOME CONNECT

Interview your family about one everyday problem. Brainstorm 5 solutions together!

🔢 Self-assessment: tick one.

I know it well!

I know some.

I need practice.

EXERCISE: AI Solving Real Problems

Name: _____ Date: _____

A. Tick (✓) the correct answer.

1. The first stage of design thinking is to ____.

(a) empathise ☐ **(b)** deploy ☐ **(c)** evaluate ☐

2. Generating many ideas is to ____.

(a) define ☐ **(b)** ideate ☐ **(c)** iterate only ☐

3. A quick, rough model of a solution is a ____.

(a) patent ☐ **(b)** poster ☐ **(c)** prototype ☐

4. Design thinking puts ____ first.

(a) people ☐ **(b)** profit ☐ **(c)** technology ☐

B. Fill in the blanks. Use the word box.

test **user** **paper** **tool**

1. Empathise, define, ideate, prototype, ____.

2. A problem statement names the ____, need and insight.

3. Prototypes can be made of ____.

4. AI is a ____, not the goal.

C. Write T for True and F for False.

1. Talking to users is part of empathy. [ ]

2. Prototypes must be perfect. [ ]

3. Testing gives feedback to improve. [ ]

4. Technology should be chosen before understanding the problem. [ ]

D. Match the following. Write the letter.

Column A		Column B
1. Empathise	____	**(a)** Generate ideas
2. Define	____	**(b)** Understand users
3. Ideate	____	**(c)** Build a quick model
4. Prototype	____	**(d)** State the problem

E. Answer in one word.

1. What is the first design-thinking step? _____

2. What is a quick model called? _____

3. What is the last step? _____

F. Answer these questions.

1. List and explain the five stages of design thinking.

2. Why should we empathise before building?

G. Case-based questions. Read and answer.

Kabir spent a day with his Paati, discovered she couldn't read small text but loved her radio, and prototyped a Tamil voice reminder. Testing showed it was too fast, so they slowed it.

1. Spending a day with Paati was _____.

(a) empathising ☐ **(b)** deploying ☐ **(c)** coding ☐

2. The voice reminder was a _____.

(a) problem ☐ **(b)** prototype ☐ **(c)** patent ☐

3. Slowing the voice came from _____.

(a) guessing ☐ **(b)** ideating ☐ **(c)** testing ☐

H. THINK DEEPER

Use design thinking to plan a solution for students who forget homework. Where might AI help, and where would it not?

UNIT REVIEW: AI SOLVE

Name: _____ Date: _____

Circle the right answer!

Q1. Machines doing tasks is...

a) Automation

b) Art

c) Rest

Q2. Design thinking starts with...

a) Code

b) Empathy

c) Money

Q3. A first test model is a...

a) Cake

b) Trophy

c) Prototype

Q4. AI helping humans is...

a) Augmentation

b) Destruction

c) Hiding

Q5. Factories use AI to spot...

a) Rainbows

b) Defects

c) Birds

Q6. Generating ideas is to...

a) Sleep

b) Delete

c) Ideate

WORD SEARCH: AI SOLVE

🔍 **Find these words. They go across → or down ↓.**

AUTOMATION	INDUSTRY	EMPATHY	PROTOTYPE	IDEATE	DESIGN

E S N A C F U B K I
A I R W N O N I U N
P D E S I G N D E D
M D B E R G I E O U
K O S N U M T A U S
O U H L A I W T F T
L T U U D G U E I R
K D E M P A T H Y Y
A U T O M A T I O N
C P R O T O T Y P E

🏅 **I finished AI SOLVE! Teacher's sign: _____**

MONTH 4

AI RISE

Skills • Careers • Future

Area: AI & Career

In this unit, you will explore:

Topic 7

Your Road to an AI Career

Topic 8

AI Skills Beyond School

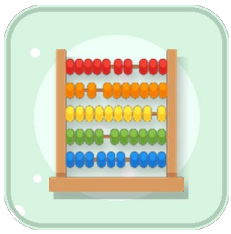
TOPIC 7 • MONTH 4 • AI & CAREER

Your Road to an AI Career

LET'S LEARN

- After Class 10, choose a stream that fits your goals.
- AI careers need maths, coding or data skills.
- Many roads: degrees, diplomas, online courses!

Roads to AI



Maths

Computers

Data

Design

MAGIC WORDS

Degree University qualification

Certificate Short course proof

Entrance exam Admission test

💬 TECH TALK: Your Road to an AI Career

💬 **GAIO Squad** Riya, Kabir, Bolt • online

👧 **Riya**

Bolt, after Class 10, which stream leads to AI 🤔?

10:02

🤖 **Bolt**

Science with maths is the most direct route 📐. But commerce and humanities also connect to AI.

10:04 ✓✓

👦 **Kabir**

How?

10:06

🤖 **Bolt**

Commerce + data business analytics 📊. Humanities + AI policy, ethics, linguistics ⚖️.

10:08 ✓✓

👧 **Riya**

What after Class 12?

10:10

🤖 **Bolt**

Degrees like B.Tech, B.Sc in data science, BCA 🎓. Entrance exams decide many admissions 📝.

10:12 ✓✓

Kabir
Any shortcuts?
10:14

Bolt
Not shortcuts, but extra roads: diplomas, certificates and online courses 💻 add skills.
10:16 ✓✓

Riya
Choose a stream that fits your goals, not just friends 🧭!
10:18

😊 Type a message...

📌 **Pinned by Bolt**

After Class 10, choose a stream that fits your goals. AI careers need maths, coding or data skills, reached through degrees, diplomas, certificates and online courses.

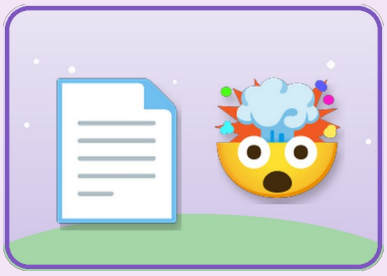
🔄 How it works

INDIA SPOTLIGHT

Many Indian universities and IITs now offer degrees in AI and data science, and admissions often use national entrance exams.

AI CHRONICLES: Two Roads, One Dream

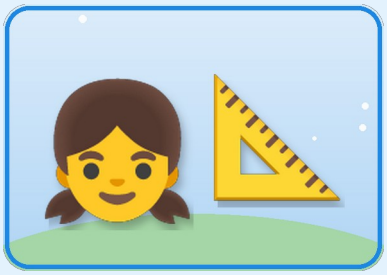
Riya Kabir Counsellor Ms. Grace



SCENE 1

Class 10 results were coming. Riya and Kabir both dreamed of AI careers, but their strengths differed.

Riya loved maths and physics. Ms. Grace suggested science with maths, aiming for B.Tech in AI 🎓.

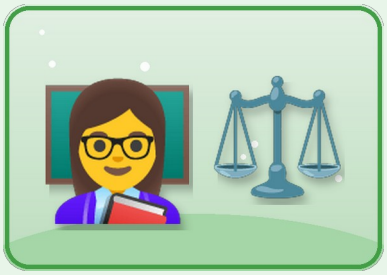


SCENE 2

SCENE 3

Kabir loved economics and debating. "Is AI closed to me?" he worried 😟.

Ms. Grace showed him AI policy, business analytics and tech journalism. "AI needs your skills too!"



SCENE 4

SCENE 5

Both enrolled in a free online Python course alongside school. They studied together 🤝.

Different streams, one shared dream. "There are many roads to AI," Kabir smiled.

SCENE 6

🔮 What happens next? Tick ✓ one!

☐	☐	☐
They co-create a project combining both skills	**Kabir drops his dream**	**The roads merge into a highway**

🧐 Story check: answer the questions.

1. Which stream suited Riya, and why?

2. What AI careers fit Kabir?

3. What did both do beyond school?

Rate this story: shade the stars.

☆ ☆ ☆ ☆ ☆

LOOK AROUND YOU: Pathways after school

B.Tech / B.E. in CS & AI

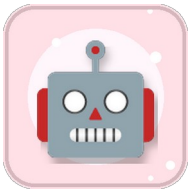
B.Sc. in Data Science

BCA / BBA Analytics

B.Des in UX / Interaction

Diploma & ITI tech courses

Online certificates



BOLT SAYS:
Your stream is a starting point, not a cage. Many roads lead to AI!

FUN FACT!
Many Indian universities now offer AI and Data Science degrees and minors!

👀 **I spy AI! Draw one AI helper you saw today.**

LET'S DO IT! Career Road Map

You need:

Road map sheet

Colours

College brochures

Steps:

1. Pick one AI-related career you like.
2. Find: stream, entrance exam, degree, skills.
3. Draw the road from Class 10 to that career.
4. Add 2 "detours": backup routes if plans change.

THINK & TALK

What is your Plan B? Is it still exciting?

Draw what you did!

WORKSHEET: Your Road to an AI Career (Part A)

Name: _____ Date: _____

1. Match the course to its focus.

B.Tech CS & AI

B.Sc Data Science

B.Des UX

BBA Analytics

Business data

User experience

Building AI systems

Statistics & data

2. Fill in the blanks. Use the word box.

WORD BOX: certificate ✿ degree ✿ entrance exam

A university qualification is a _____.

Proof of a short course is a _____.

An admission test is an _____.

WORKSHEET: Your Road to an AI Career (Part B)

Name: _____ Date: _____

1. Can this road lead to AI work? Tick ✓.

	Yes	Unlikely
B.Tech CS		
B.Sc Data Science		
B.Des UX		
Online AI certificate		

2. Draw your road map from Class 10 to your dream career!

Plan A: _____ Plan B: _____

PUZZLE FUN: Your Road to an AI Career

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.

A				
B				
C				

2. Define it! Write each term in your own words with an example.

Degree

Meaning: _____

Example: _____

Certificate

Meaning: _____

Example: _____

Entrance exam

Meaning: _____

Example: _____

🧪 DEMO LAB: Career roadmap research (online, with teacher)

Name: _____ Date: _____

🎯 **Goal:** Plan a real road from Class 10 to an AI-related career.

🔗 **Open:** university / college websites + career portals | 👩‍🏫 **With a teacher or parent**

👣 Steps

1		Pick one AI-related career (e.g., data scientist, AI policy analyst).
2		Find which Class 11 stream and subjects suit it.
3		Find 2 entrance exams or admission routes.
4		List 3 colleges/courses and 1 online certificate.
5		Draw a timeline: Class 10 → 12 → degree → first job.

🔬 Record your observations

Career chosen:	
Entrance route:	
Online certificate found:	

💭 **Reflect:** What's your backup road if plan A doesn't work out?

? QUIZ TIME: Your Road to an AI Career

Circle the right answer!

Q1. Many AI careers need...

a) Maths & CS

b) Only cooking

c) Only singing

Q2. An admission test is an...

a) Party

b) Entrance exam

c) Event

Q3. UX design focuses on...

a) Tractors

b) Volcanoes

c) User experience

Q4. Is your stream a cage?

a) No, a start

b) Yes, forever

c) Maybe

Q5. A short course gives a...

a) House

b) Certificate

c) Pizza

MY SCORE: ____ / 5. Shade one star per correct answer.

☆ ☆ ☆ ☆ ☆

✅ TRUE OR FALSE: Your Road to an AI Career

Look at the picture. Tick ✓ TRUE or FALSE.

		👍 TRUE	👎 FALSE
	Many AI careers need maths and CS.		
	Your stream locks you forever.		
	Design degrees can lead to AI-related work.		
	Entrance exams are admission tests.		
	Having a Plan B is wise.		
	Online certificates are worthless.		

HOME CONNECT

Discuss your stream choice with your family. Draw Plan A and Plan B together.

🔢 Self-assessment: tick one.

I know it well!

I know some.

I need practice.

EXERCISE: Your Road to an AI Career

Name: _____ Date: _____

A. Tick (✓) the correct answer.

1. The most direct stream for AI engineering is ____.

(a) science with maths ☐ **(b)** no stream ☐ **(c)** only arts ☐

2. Exams for college admission are ____ exams.

(a) unit ☐ **(b)** entrance ☐ **(c)** practice ☐

3. A short course with a certificate is a ____.

(a) degree ☐ **(b)** debate ☐ **(c)** certificate course ☐

4. Humanities + AI can lead to AI ____.

(a) policy ☐ **(b)** plumbing ☐ **(c)** piloting ☐

B. Fill in the blanks. Use the word box.

goals **degrees** **analytics** **courses**

1. Choose a stream that fits your ____.

2. B.Tech and B.Sc are ____.

3. Commerce + data leads to business ____.

4. Online ____ add extra skills.

C. Write T for True and F for False.

1. There are many roads to AI careers. [ ]

2. Only science students can work in AI. [ ]

3. Entrance exams decide many admissions. [ ]

4. Online courses can add useful skills. [ ]

D. Match the following. Write the letter.

Column A		Column B
1. Degree	____	(a) Admission test
2. Diploma	____	(b) University qualification
3. Entrance exam	____	(c) Proof of a short course
4. Certificate	____	(d) Practical qualification

E. Answer in one word.

1. What exam decides many admissions? _____
2. Name one AI-related degree. _____
3. Commerce + data leads to? _____

F. Answer these questions.

1. Describe two different roads to an AI career after Class 10.

2. Why should you choose a stream based on goals, not friends?

G. Case-based questions. Read and answer.

Riya, strong in maths, chose science aiming for B.Tech AI. Kabir, strong in economics and debate, discovered AI policy and analytics. Both took a free online Python course.

1. Riya's choice suits _____.

(a) AI engineering ☐ **(b)** dance ☐ **(c)** cooking ☐

2. Kabir's path shows AI also needs _____.

(a) only coders ☐ **(b)** policy and analysis skills ☐ **(c)** no skills ☐

3. The online course added _____.

(a) stress only ☐ **(b)** nothing ☐ **(c)** extra skills ☐

H. THINK DEEPER

Map three different AI careers to three different Class 11 streams. Explain the link for each.

TOPIC 8 • MONTH 4 • AI & CAREER

AI Skills Beyond School

LET'S LEARN

- Real-world AI skills grow outside the classroom too.
- Online courses, clubs, internships and projects all help.
- Communication and ethics matter as much as code!

Beyond-school skills

Online courses

Competitions

Projects

Communication

MAGIC WORDS

Portfolio
Showcase of work

Networking
Building connections

Internship Short work training

💬 TECH TALK: AI Skills Beyond School

💬 GAIO Squad Riya, Kabir, Bolt • online

👦 Kabir

Bolt, my marks are good. Is that enough for AI jobs 📄?

10:02

🤖 Bolt

Marks open doors; skills and proof keep them open. Build a portfolio 📁 of real projects.

10:04 ✓✓

👧 Riya

Where do we learn beyond school?

10:06

🤖 Bolt

Online courses 💻, coding clubs, competitions 🏆, hackathons and open-source projects.

10:08 ✓✓

👦 Kabir

What's networking 🤝?

10:10

🤖 Bolt

Building respectful connections with mentors and peers: at events, clubs, or safe professional platforms.

10:12 ✓✓

Riya
And internships?
10:14

Bolt
Short work experiences where you learn real tools and teamwork. Many are open after Class 12.
10:16 ✓✓

Kabir
Code, communicate, act ethically ✅!
10:18

Type a message...

Pinned by Bolt

AI skills grow outside class through courses, clubs, competitions and projects. Build a portfolio, network respectfully and seek internships. Communication and ethics matter as much as code.

How it works

INDIA SPOTLIGHT

National platforms like SWAYAM and NPTEL offer free online courses, including AI and data science, often taught by IIT professors.

📚 AI CHRONICLES: The Portfolio That Spoke

👦 Kabir

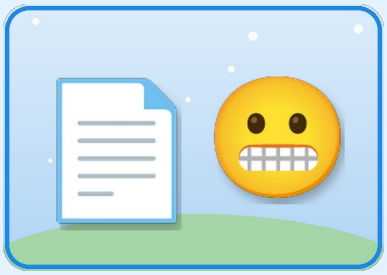
👩 Ms. Neha (start-up founder)

🤖 Bolt

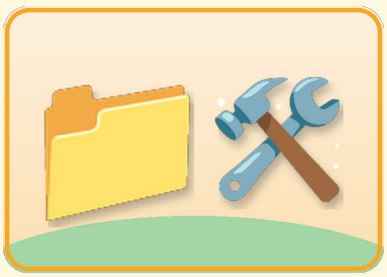
SCENE 1

A local AI start-up announced a summer programme for Class 10–12 students ☀️.

Kabir had good marks, but so did 300 other applicants.



SCENE 2

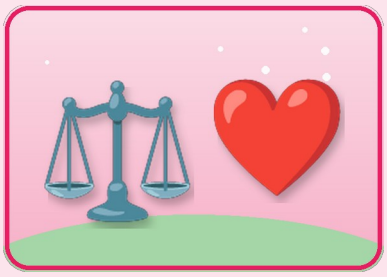


SCENE 3

His portfolio stood out: a plant-disease classifier, a hackathon app and a blog explaining them.

In the interview, Ms. Neha asked him to explain a mistake. He described how he fixed overfitting 🔁.

SCENE 4



SCENE 5

She also asked about data privacy. Kabir explained how he got consent for photos 📸.

He was selected. "Skills, clear communication and ethics: that's what we hire," said Ms. Neha.

SCENE 6

🔮 What happens next? Tick ✓ one!

☐	☐	☐
Kabir mentors juniors next summer	**Kabir stops building projects**	**The start-up hires Bolt instead**

🧐 Story check: answer the questions.

1. What made Kabir stand out?

2. What did he explain in the interview?

3. Which three qualities did Ms. Neha value?

Rate this story: shade the stars.

☆ ☆ ☆ ☆ ☆

LOOK AROUND YOU: Ways to grow your skills

Free coding platforms

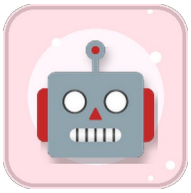
Olympiads & hackathons

Science clubs

GitHub-style portfolio

Debate & MUN

Volunteering with tech



BOLT SAYS:
Build things! A small finished project teaches more than ten unfinished ones.

FUN FACT!
Many top universities share free online courses on AI and programming!

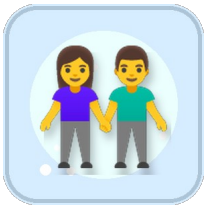
👀 **I spy AI! Draw one AI helper you saw today.**

LET'S DO IT! My Skill Sprint

You need:

4-week planner

Online course (free)



Study buddy

Steps:

1. Pick one skill: Python, design or public speaking.
2. Plan 20 minutes a day for 4 weeks.
3. Check in with your study buddy weekly.
4. Add your result to your portfolio!

THINK & TALK

What helped you stay consistent?

Draw what you did!

WORKSHEET: AI Skills Beyond School (Part A)

Name: _____ Date: _____

1. Circle the ways to build skills beyond school.

Online course

Endless TV

Hackathon

Personal project

Skipping practice

Debate club

2. Fill in the blanks. Use the word box.

WORD BOX: internship ✿ portfolio ✿ networking

A showcase of your work is a _____.

Building connections is _____.

Short work training is an _____.

WORKSHEET: AI Skills Beyond School (Part B)

Name: _____ Date: _____

1. Match the activity to the skill it builds.

Debate club ●

Hackathon ●

Volunteering ●

Online course ●

●

Problem solving

●

Empathy

●

Technical knowledge

●

Speaking

2. Rate yourself! Tick ✓ for you.

	Strong	Growing
Coding		
Communication		
Ethics		
Teamwork		

PUZZLE FUN: AI Skills Beyond School

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.

A				
B				
C				

2. Define it! Write each term in your own words with an example.

Portfolio

Meaning: _____

Example: _____

Networking

Meaning: _____

Example: _____

Internship

Meaning: _____

Example: _____

🧪 DEMO LAB: SWAYAM / NPTEL + portfolio page (with teacher)

Name: _____ Date: _____

🎯 **Goal:** Find a course and design your skills portfolio.
🔗 **Open:** swayam.gov.in / nptel.ac.in | 👩‍🏫 **With a teacher or parent**

👣 Steps

1. Browse SWAYAM/NPTEL for an AI, data or Python course.
2. Note duration, level and certificate details.
3. Design a one-page portfolio: about me, 3 projects, skills.
4. Write a polite message to a mentor asking for advice (don't send without a parent/teacher).
5. Find one internship or competition open to school students.

🔬 Record your observations

Course found:	
Projects in my portfolio:	
Competition/internship found:	

💭 **Reflect:** Why are communication and ethics as important as coding in AI careers?

QUIZ TIME: AI Skills Beyond School

Circle the right answer!

Q1. A showcase of your work is a...

a) Portfolio

b) Bag

c) Tiffin

Q2. Short work training is an...

a) Inflation

b) Internship

c) Interlude

Q3. Besides code, AI careers need...

a) Sleep only

b) Nothing

c) Communication & ethics

Q4. Better to finish...

a) 1 small project

b) 10 half projects

c) None

Q5. Building connections is...

a) Fishing

b) Networking

c) Knitting

MY SCORE: ____ / 5. Shade one star per correct answer.

☆ ☆ ☆ ☆ ☆

✅ TRUE OR FALSE: AI Skills Beyond School

Look at the picture. Tick ✓ TRUE or FALSE.

		👍 TRUE	👎 FALSE
🌐	Free online courses can teach AI.		
🗂️	A portfolio showcases your work.		
🙈	Communication doesn't matter in tech.		
🏅	An internship is short work training.		
🔟	Ten unfinished projects beat one finished one.		
🤝	Networking means building connections.		

HOME CONNECT

Choose a 4-week skill sprint and ask a family member to be your check-in buddy.

🔢 Self-assessment: tick one.

I know it well!

I know some.

I need practice.

EXERCISE: AI Skills Beyond School

Name: _____ Date: _____

A. Tick (✓) the correct answer.

1. A collection of your real projects is a ____.

(a) portfolio ☐ **(b)** passport ☐ **(c)** password ☐

2. Building respectful professional connections is ____.

(a) gaming ☐ **(b)** networking ☐ **(c)** hacking ☐

3. Short real-world work experience is an ____.

(a) invoice ☐ **(b)** interview ☐ **(c)** internship ☐

4. Free Indian online course platforms include ____.

(a) SWAYAM ☐ **(b)** Netflix ☐ **(c)** none ☐

B. Fill in the blanks. Use the word box.

skills **communication** **competitions** **code**

1. Marks open doors; ____ keep them open.

2. Explaining work clearly is ____.

3. Hackathons and ____ build skills.

4. Ethics matter as much as ____.

C. Write T for True and F for False.

1. Projects can prove your skills. [ ]

2. Only marks matter for AI jobs. [ ]

3. Networking should be respectful and safe. [ ]

4. Ethics are irrelevant in AI work. [ ]

D. Match the following. Write the letter.

Column A		Column B
1. Portfolio	____	(a) Work experience
2. Networking	____	(b) Proof of projects
3. Internship	____	(c) Coding competition
4. Hackathon	____	(d) Professional connections

E. Answer in one word.

1. What shows your projects? _____
2. What is short work experience? _____
3. Name a free national course platform. _____

F. Answer these questions.

1. List four ways to build AI skills outside school.

2. Why do employers value communication and ethics?

G. Case-based questions. Read and answer.

Among 300 applicants, Kabir stood out with a portfolio of projects, clearly explained how he fixed overfitting and described getting consent for photos.

1. Kabir's projects formed a ____.

(a) portfolio ☐ **(b)** complaint ☐ **(c)** password ☐

2. Explaining his fix showed ____.

(a) silence ☐ **(b)** communication ☐ **(c)** copying ☐

3. Getting consent showed ____.

(a) speed ☐ **(b)** luck ☐ **(c)** ethics ☐

H. THINK DEEPER

Design your own one-page portfolio plan: three projects you could build in the next year and the skill each shows.

UNIT REVIEW: AI RISE

Name: _____ Date: _____

Circle the right answer!

Q1. A showcase of your work:

a) Portfolio

b) Bag

c) Tiffin

Q2. An admission test is an...

a) Party

b) Entrance exam

c) Event

Q3. Short work training is an...

a) Inflation

b) Interlude

c) Internship

Q4. Is your stream a cage?

a) No, a start

b) Yes

c) Maybe

Q5. Building connections is...

a) Fishing

b) Networking

c) Knitting

Q6. UX design focuses on...

a) Tractors

b) Volcanoes

c) User experience

WORD SEARCH: AI RISE

🔍 **Find these words. They go across → or down ↓.**

DEGREE

PORTFOLIO

NETWORKING

INTERNSHIP

CAREER

SKILL

K	O	D	E	G	R	E	E	P	F
N	E	T	W	O	R	K	I	N	G
H	C	A	R	E	E	R	K	I	B
C	B	P	I	S	T	R	L	E	G
C	O	G	P	I	F	M	O	U	L
T	P	O	R	T	F	O	L	I	O
G	L	D	B	S	K	I	L	L	H
I	U	W	H	D	L	F	K	P	A
I	N	T	E	R	N	S	H	I	P
B	M	C	K	L	A	L	K	L	E

I finished AI RISE! Teacher's sign: _____

MONTH 5

AI CREATE

Try • Build • Innovate

Area: AI Tools

In this unit, you will explore:

Topic 9

AI Productivity Booster

Topic 10

Create Your AI Project

TOPIC 9 • MONTH 5 • AI TOOLS

AI Productivity Booster

LET'S LEARN

- AI can help you plan, summarise and organise.
- Use it to save time on routine tasks.
- Then spend that time on deep learning and rest!

Productivity tools

Planning

Summarising

To-do lists

Focus timers

MAGIC WORDS

Pomodoro Focus-break method

Priority Most important task

Productivity Getting more done well

💬 TECH TALK: AI Productivity Booster

💬 **GAIO Squad** Riya, Kabir, Bolt • online

👧 **Riya**
Board exams + projects + tuition. I'm drowning 😵!
10:02

🤖 **Bolt**
Let's plan. First, prioritise 📋: urgent and important first, unimportant later.
10:04 ✓✓

👦 **Kabir**
Can AI help?
10:06

🤖 **Bolt**
Yes! Ask it to turn your syllabus into a weekly timetable 📅 or summarise long notes.
10:08 ✓✓

👧 **Riya**
I get distracted every 10 minutes 📱.
10:10

🤖 **Bolt**
Try Pomodoro 🍅: 25 minutes focused study, 5 minutes break. Phone away!
10:12 ✓✓

👦 **Kabir**

So AI saves time on routine tasks?

10:14

🤖 **Bolt**

Exactly. Use the saved time for deep practice, sleep 😴 and exercise. Productivity isn't just doing more.

10:16 ✓✓

👧 **Riya**

Plan smart, focus deep, rest well ✅!

10:18

😊 Type a message...

📌 **Pinned by Bolt**

AI can plan, summarise and organise routine work. Prioritise tasks, use focus methods like Pomodoro and use saved time for deep learning and rest.

🔄 How it works

INDIA SPOTLIGHT

Many Indian students prepare for board exams with digital study planners and free national e-learning resources.

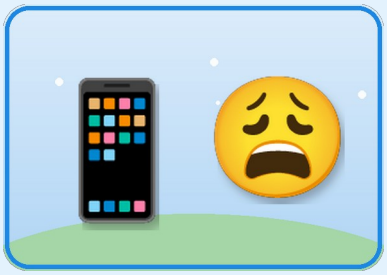
AI CHRONICLES: The Night Owl's Reset

Riya | Mum | Study Planner AI

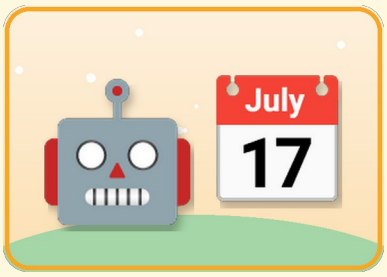
SCENE 1

Riya studied till 2 a.m. every night before boards, yet felt she remembered nothing 😵.

Her phone buzzed every few minutes. 'Quick checks' ate two hours a day.



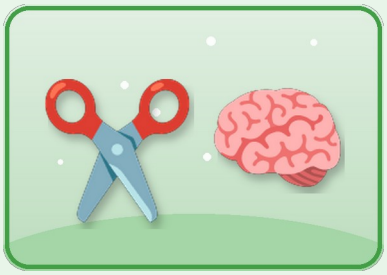
SCENE 2



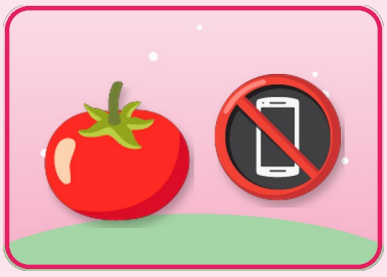
SCENE 3

She asked an AI planner to build a timetable with priorities, breaks and 8 hours of sleep.

She edited it for tuition timings and her weak subjects. AI didn't know those!



SCENE 4



SCENE 5

Pomodoro rounds with the phone in Mum's room. Focus improved 🎯.

Rested and organised, her mock scores rose by 15%. "Smart planning beats late nights," said Mum.

SCENE 6

🔮 What happens next? Tick ✓ one!

☐	☐	☐
Riya shares her planner with friends	**Riya returns to 2 a.m. study**	**The phone joins Pomodoro**

🧐 Story check: answer the questions.

1. Why wasn't late-night studying working?

2. What did Riya change in the AI's plan?

3. How did Pomodoro help?

Rate this story: shade the stars.

☆ ☆ ☆ ☆ ☆

LOOK AROUND YOU: Smart productivity habits

AI study timetable

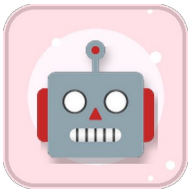
Chapter summaries to revise

Practice question sets

25-min focus + 5-min break

Organised notes

Sleep 8 hours



BOLT SAYS:
I can make your timetable. Only you can follow it!

FUN FACT!
The Pomodoro technique uses 25 minutes of focus followed by a short break!

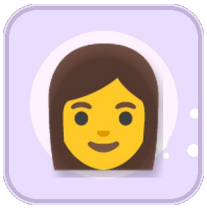
I spy AI! Draw one AI helper you saw today.

LET'S DO IT! Board Exam Planner

You need:

Planner sheet

Syllabus list



Teacher's AI tool

Steps:

1		List all subjects and chapters left.
2		Ask AI (with teacher) for a 4-week plan with breaks.
3		Edit the plan to fit YOUR routine.
4		Try 2 Pomodoros today!

THINK & TALK

Which part of the AI plan did you change? Why?

Draw what you did!

WORKSHEET: AI Productivity Booster (Part A)

Name: _____ Date: _____

1. Smart study session. Write 1, 2, 3, 4.

Take a short break ☐

Choose top priority ☐

Review & summarise ☐

Focus 25 minutes ☐

2. Productive or not? Tick ✓.

		Productive	Not
	25-min focus sessions		
	Phone notifications on		
	Weekly plan		
	Studying all night daily		
	Ticking a to-do list		

WORKSHEET: AI Productivity Booster (Part B)

Name: _____ Date: _____

1. Match the tool to the job.

Planner ●

Timer ●

To-do list ●

Summariser ●

●

Quick revision

●

Tracking tasks

●

Weekly schedule

●

Focus sessions

2. Fill in the blanks. Use the word box.

WORD BOX: priority ✿ Pomodoro ✿ productivity

25-min focus + break is the _____ method.

The most important task is the _____.

Getting more done well is _____.

PUZZLE FUN: AI Productivity Booster

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.

A				
B				
C				

2. Define it! Write each term in your own words with an example.

Pomodoro

Meaning: _____

Example: _____

Priority

Meaning: _____

Example: _____

Productivity

Meaning: _____

Example: _____

🧪 DEMO LAB: A school-approved AI chatbot + timer (with teacher)

Name: _____ Date: _____

🎯 **Goal:** Build an AI-assisted study plan and test the Pomodoro method.

🔗 **Open:** any approved chatbot + phone timer | 👩‍🏫 **With a teacher or parent**

👣 Steps

1		List all subjects and pending tasks. Mark each Urgent/Important.
2		Ask AI: 'Make a 7-day study plan for these, with breaks and 8 hours sleep.'
3		Edit the plan: AI doesn't know your tuition timings!
4		Do 2 Pomodoro rounds (25 + 5 minutes) with the phone in another room.
5		Ask AI to summarise one chapter; check it against your textbook.

🔬 Record your observations

Top priority task:	
Changes I made to AI's plan:	
Focus level in Pomodoro (1–5):	

💭 **Reflect:** How can AI save you time without making you less capable?

? QUIZ TIME: AI Productivity Booster

Circle the right answer!

Q1. 25 min focus + break is...

a) Pomodoro

b) Pizza

c) Gaming

Q2. The most important task is the...

a) Balloon

b) Priority

c) Rubbish

Q3. AI can make your timetable. Who follows it?

a) AI

b) TV

c) You

Q4. Productive students also...

a) Sleep well

b) Never sleep

c) Scroll all day

Q5. AI summaries help with...

a) Cooking

b) Revision

c) Painting

MY SCORE: _____ / 5. Shade one star per correct answer.

☆ ☆ ☆ ☆ ☆

✅ TRUE OR FALSE: AI Productivity Booster

Look at the picture. Tick ✓ TRUE or FALSE.

	TRUE	FALSE
Pomodoro uses focus sessions with breaks.		
AI can help make a study plan.		
Skipping sleep boosts exam scores.		
Do the top priority first.		
Notifications help you focus.		
Summaries help with revision.		

HOME CONNECT

Try one Pomodoro session with a family member for a household task!

🔢 Self-assessment: tick one.

I know it well!

I know some.

I need practice.

EXERCISE: AI Productivity Booster

Name: _____ Date: _____

A. Tick (✓) the correct answer.

1. 25 minutes focus + 5 minutes break is the ____ method.

(a) Pomodoro ☐ **(b)** Pythagoras ☐ **(c)** Pareto-only ☐

2. Deciding what to do first is setting ____.

(a) passwords ☐ **(b)** priorities ☐ **(c)** pixels ☐

3. Getting more useful work done well is ____.

(a) procrastination ☐ **(b)** panic ☐ **(c)** productivity ☐

4. AI is best used for ____ tasks like planning.

(a) routine ☐ **(b)** sleeping ☐ **(c)** eating ☐

B. Fill in the blanks. Use the word box.

important **summarise** **away** **sleep**

1. Urgent and ____ tasks come first.

2. AI can ____ long notes.

3. Keep your phone ____ while studying.

4. Use saved time for ____ and exercise.

C. Write T for True and F for False.

1. AI can turn a syllabus into a timetable. [ ]

2. Productivity means studying 18 hours a day. [ ]

3. AI plans may need human editing. [ ]

4. Sleep harms exam performance. [ ]

D. Match the following. Write the letter.

Column A		Column B
1. Pomodoro	____	**(a)** Short version
2. Priority	____	**(b)** Focus + break cycles
3. Summary	____	**(c)** Recharges the brain
4. Rest	____	**(d)** Order of importance

E. Answer in one word.

1. Which method uses 25-minute focus blocks? _____

2. What do we set to order tasks? _____

3. What should AI summaries be checked against? _____

F. Answer these questions.

1. Explain how AI can improve productivity without reducing learning.

2. Describe the Pomodoro method.

G. Case-based questions. Read and answer.

Riya studied till 2 a.m. with constant phone checks. An AI planner made a timetable; she edited it for tuition timings and used Pomodoro. Her mock scores rose 15%.

1. Phone checks were a _____.

(a) distraction ☐ **(b)** priority ☐ **(c)** summary ☐

2. Editing the plan shows that _____.

(a) AI knows everything ☐ **(b)** AI plans need human context ☐ **(c)** plans are useless ☐

3. Her improvement came from _____.

(a) less sleep ☐ **(b)** more phone use ☐ **(c)** planning, focus and rest ☐

H. THINK DEEPER

Design an ideal AI-assisted study week for board exams. Where would you deliberately NOT use AI, and why?

TOPIC 10 • MONTH 5 • AI TOOLS

Create Your AI Project

LET'S LEARN

- An AI project follows a clear life cycle.
- Problem → Data → Model → Evaluate → Deploy.
- Start small, document everything and present clearly!

AI project cycle

Problem scoping

Data

Model

Deploy

MAGIC WORDS

Scoping Defining the problem

Evaluate Measure performance

Deploy Put into real use

💬 TECH TALK: Create Your AI Project

💬 GAIO Squad Riya, Kabir, Bolt • online

👦 Kabir
Bolt, I want to build an AI that detects plant disease 🌿!
10:02

🤖 Bolt
Great! First scope it 🎯: which plant, which diseases, who will use it?
10:04 ✓✓

👧 Riya
Tomato leaves, 3 diseases, for local farmers.
10:06

🤖 Bolt
Next, data 📸: hundreds of labelled leaf photos. Then train a model 🧠.
10:08 ✓✓

👦 Kabir
How do we know it works?
10:10

🤖 Bolt
Evaluate 📏 on a test set: accuracy, plus where it makes mistakes.
10:12 ✓✓

👧 **Riya**

And then?

10:14

🤖 **Bolt**

Deploy 🚀: put it in a simple app or webpage. Document every step 📒 and present clearly.

10:16 ✓✓

👦 **Kabir**

Problem data model evaluate deploy 🔁!

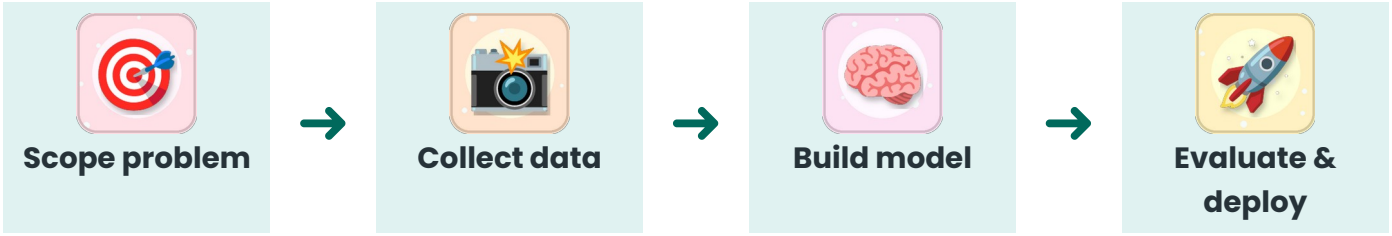
10:18

😊 Type a message...

📌 **Pinned by Bolt**

An AI project follows a life cycle: scope the problem, collect data, build a model, evaluate it and deploy. Start small, document everything and present clearly.

🔄 How it works



INDIA SPOTLIGHT

CBSE's AI curriculum includes an AI project cycle, and many Indian students showcase AI projects at science and innovation fairs.

AI CHRONICLES: The Leaf Doctor

Kabir

Murugan Anna (farmer)

Riya

SCENE 1

Murugan Anna's tomato crop was dying, and he couldn't tell which disease it was 😟.

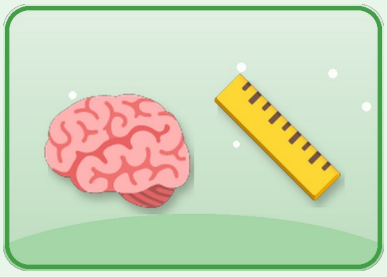
Kabir and Riya scoped a project: detect 3 common tomato leaf diseases from phone photos.

SCENE 2

SCENE 3

They collected 600 labelled leaf photos with help from an agriculture officer.

Their model scored 88% on the test set but confused two diseases. They added more photos of both.


SCENE 4

SCENE 5

They deployed it as a simple webpage. Murugan Anna snapped a leaf and got an answer in Tamil.

Their documented project won the district science fair. "Start small, solve real," said Riya.

SCENE 6

🔮 What happens next? Tick ✓ one!

☐	☐	☐
They add advice on treatment	**They skip testing next time**	**The tomatoes write a thank-you note**

🧐 Story check: answer the questions.

1. How did they scope the project?

2. What problem did evaluation reveal?

3. How was the model deployed?

Rate this story: shade the stars.

☆ ☆ ☆ ☆ ☆

LOOK AROUND YOU: Class 10 AI project ideas

Leaf disease detector

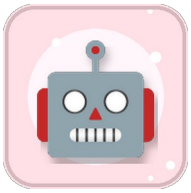
Mask detector

Waste sorter

Book recommender

Weather predictor

Voice-controlled helper



BOLT SAYS:
Many classroom AI projects use free no-code tools to train simple models from your own photos!

FUN FACT!
The AI project cycle is part of the CBSE AI curriculum for Classes 9 and 10!

I spy AI! Draw one AI helper you saw today.

LET'S DO IT! Teachable Model Lab

You need:

Webcam

Objects

No-code AI tool

Steps:

1		Problem: sort fruits automatically.
2		Capture 30 photos of each fruit.
3		Train a model on a no-code tool.
4		Test with new fruits. Record accuracy!

THINK & TALK

How could you improve your model's accuracy?

Draw what you did!

WORKSHEET: Create Your AI Project (Part A)

Name: _____ Date: _____

1. AI project cycle. Write 1, 2, 3, 4.

Deploy ☐

Problem scoping ☐

Build model ☐

Collect data ☐

2. Plan your own AI project: problem, data, model, users!

My project: _____

WORKSHEET: Create Your AI Project (Part B)

Name: _____ Date: _____

1. Match the cycle stage to the action.

Scoping ●

Data ●

Evaluate ●

Deploy ●

● Measure accuracy

● Define the problem

● Give to real users

● Collect photos

2. Fill in the blanks. Use the word box.

WORD BOX: deploy ✿ scoping ✿ evaluate

Defining the problem is _____.

To measure performance is to _____.

To put into real use is to _____.

PUZZLE FUN: Create Your AI Project

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.

A				
B				
C				

2. Define it! Write each term in your own words with an example.

Scoping

Meaning: _____

Example: _____

Evaluate

Meaning: _____

Example: _____

Deploy

Meaning: _____

Example: _____

🧪 DEMO LAB: Teachable Machine + project template (with teacher)

Name: _____ Date: _____

🎯 **Goal:** Complete a mini AI project life cycle.

🔗 **Open:** teachablemachine.withgoogle.com | 👩‍🏫 **With a teacher or parent**

👣 Steps

1		Scope: e.g., 'Classify recyclable vs non-recyclable waste'.
2		Collect 40 photos per class with your camera.
3		Train the model.
4		Evaluate on 10 new items per class. Record accuracy and errors.
5		Export/share the model link and write a 1-page project report.

🔬 Record your observations

Project scope:	
Test accuracy:	
Biggest error and why:	

💭 **Reflect:** What would you need before deploying this for real users?

? QUIZ TIME: Create Your AI Project

Circle the right answer!

Q1. Defining the problem is...

a) Scoping

b) Launching

c) Resting

Q2. Putting AI into real use is...

a) Delete

b) Deploy

c) Hide

Q3. Measuring performance is...

a) Decorate

b) Compose

c) Evaluate

Q4. The AI project cycle is in...

a) CBSE AI curriculum

b) Cookbooks

c) Games only

Q5. Start projects...

a) Huge

b) Small

c) Never

MY SCORE: _____ / 5. Shade one star per correct answer.

☆ ☆ ☆ ☆ ☆

✅ TRUE OR FALSE: Create Your AI Project

Look at the picture. Tick ✓ TRUE or FALSE.

		👍 TRUE	👎 FALSE
	Projects start with problem scoping.		
	Data quality affects the model.		
	Deploy means delete the model.		
	Evaluate means measure performance.		
	The AI project cycle is in the CBSE AI curriculum.		
	Always start with a huge project.		

HOME CONNECT

Explain your AI project idea to your family. Ask: who would use it?

🔢 Self-assessment: tick one.

I know it well!

I know some.

I need practice.

EXERCISE: Create Your AI Project

Name: _____ Date: _____

A. Tick (✓) the correct answer.

1. The first stage of an AI project is ____ the problem.

(a) scoping ☐ **(b)** deploying ☐ **(c)** selling ☐

2. Checking a model on test data is ____.

(a) collecting ☐ **(b)** evaluating ☐ **(c)** decorating ☐

3. Making the model available to users is ____.

(a) deleting ☐ **(b)** labelling ☐ **(c)** deploying ☐

4. Recording every step is ____.

(a) documentation ☐ **(b)** duplication ☐ **(c)** decoration ☐

B. Fill in the blanks. Use the word box.

deploy **labelled** **small** **mistakes**

1. Problem → data → model → evaluate → ____.

2. Training needs ____ data.

3. Start ____ and solve real problems.

4. Evaluation shows where the model makes ____.

C. Write T for True and F for False.

1. Scoping defines users and goals. [ ]

2. Evaluation is optional. [ ]

3. Documentation helps others understand your work. [ ]

4. Deploy before testing. [ ]

D. Match the following. Write the letter.

Column A		Column B
1. Scoping	____	**(a)** Test performance
2. Data	____	**(b)** Define the problem
3. Evaluate	____	**(c)** Release to users
4. Deploy	____	**(d)** Labelled examples

E. Answer in one word.

1. What is the first project stage? _____

2. What is releasing to users? _____

3. What records each step? _____

F. Answer these questions.

1. Explain the five stages of the AI project cycle.

2. Why is evaluation important before deployment?

G. Case-based questions. Read and answer.

Students scoped a tomato-disease detector, collected 600 labelled photos, got 88% test accuracy with two diseases confused, added more photos, then deployed it as a webpage.

1. Choosing 3 diseases for local farmers was ____.

(a) scoping ☐ **(b)** deploying ☐ **(c)** evaluating ☐

2. Finding the confusion happened during ____.

(a) scoping ☐ **(b)** evaluation ☐ **(c)** deployment ☐

3. The webpage was the ____ step.

(a) data ☐ **(b)** scoping ☐ **(c)** deployment ☐

H. THINK DEEPER

Scope an AI project for your school. Fill in: problem, users, data needed, how you'll evaluate and how you'll deploy.

UNIT REVIEW: AI CREATE

Name: _____ Date: _____

Circle the right answer!

Q1. 25 min focus + break is...

a) Pomodoro

b) Pizza

c) Gaming

Q2. Putting AI into real use is...

a) Delete

b) Deploy

c) Hide

Q3. The most important task is the...

a) Balloon

b) Rubbish

c) Priority

Q4. Defining the problem is...

a) Scoping

b) Launching

c) Resting

Q5. Measuring performance is...

a) Decorate

b) Evaluate

c) Compose

Q6. Who follows the AI timetable?

a) AI

b) TV

c) You

WORD SEARCH: AI CREATE

🔍 **Find these words. They go across → or down ↓.**

POMODORO

PRIORITY

SCOPING

EVALUATE

DEPLOY

PROJECT

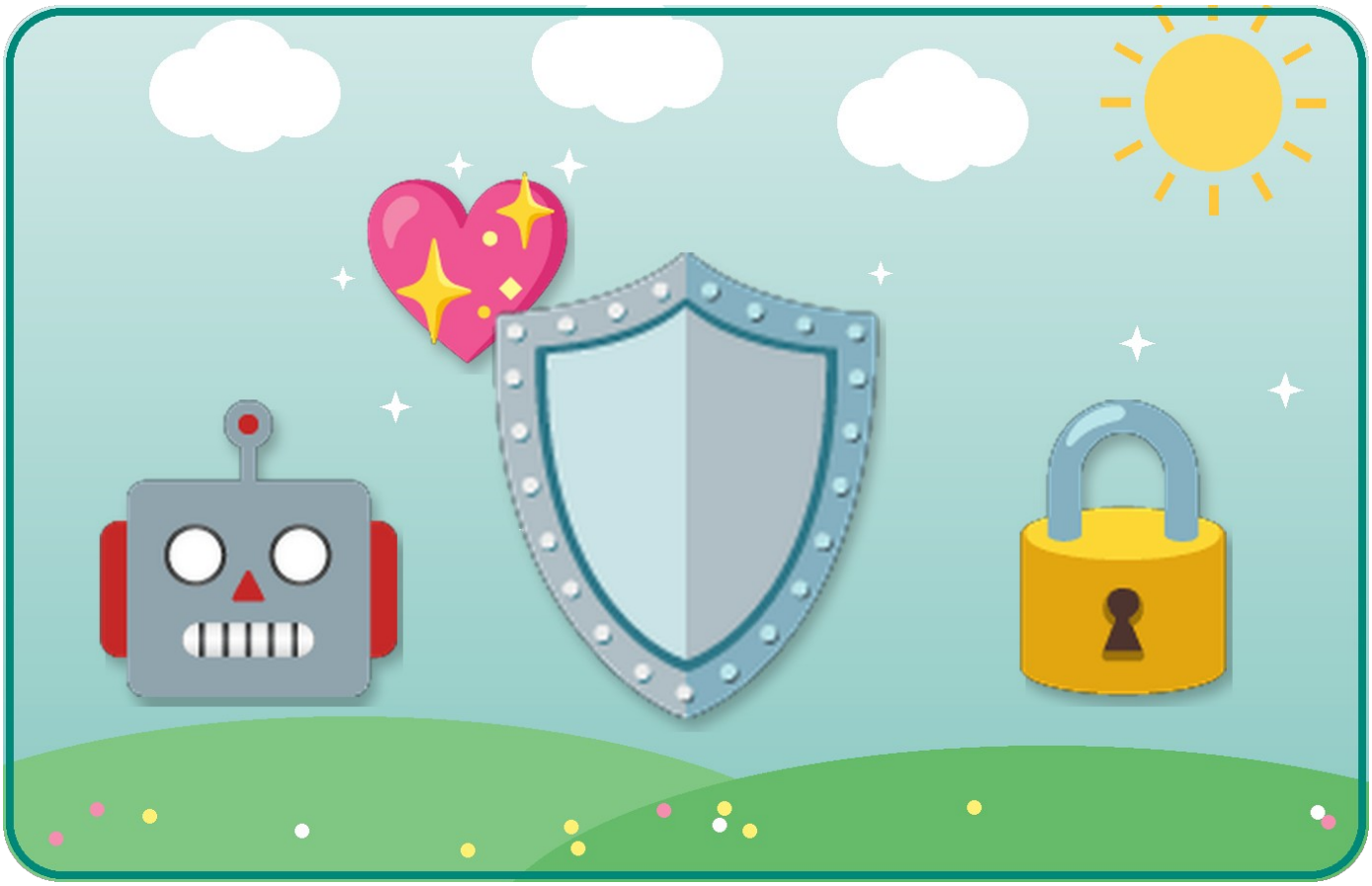
U	E	S	A	P	B	F	W	G	D
H	V	C	P	M	S	M	P	S	I
P	A	O	D	U	P	D	R	P	M
K	L	P	B	O	O	E	O	R	C
G	U	I	L	S	M	P	J	I	W
M	A	N	E	L	O	L	E	O	I
T	T	G	C	K	D	O	C	R	L
K	E	F	C	E	O	Y	T	I	K
R	F	B	C	W	R	T	N	T	B
H	W	M	I	P	O	O	E	Y	B

I finished AI CREATE! Teacher's sign: _____

MONTH 6

AI CARE

Verify • Protect • Respect



Area: Responsible AI

In this unit, you will explore:

Topic 11

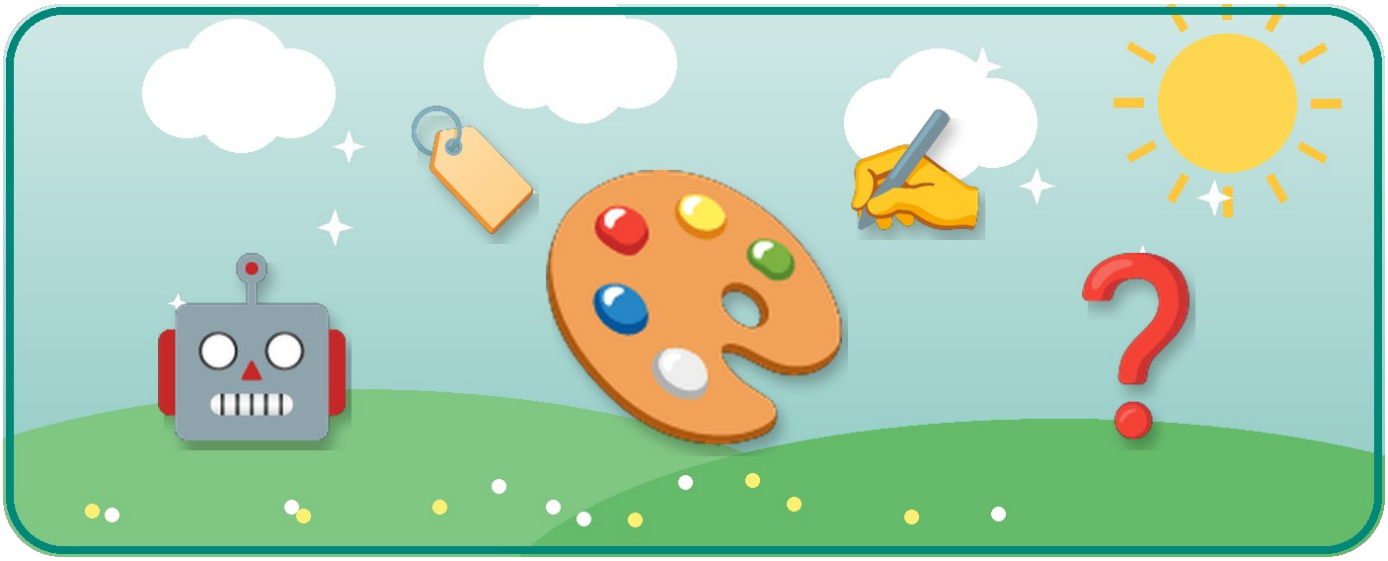
Original or AI-Made?

Topic 12

Use AI, Don't Misuse AI

TOPIC 11 • MONTH 6 • RESPONSIBLE AI

Original or AI-Made?



LET'S LEARN

- AI can write essays, make art and compose music.
- Passing off AI work as your own is plagiarism.
- Disclose AI use, cite sources and add your own thinking!

Honest creating

Disclose AI use

Cite sources

Add your ideas

Your own words

MAGIC WORDS

Plagiarism Using others' work as yours

Disclosure Saying AI helped

Watermark Hidden ID mark

💬 TECH TALK: Original or AI-Made?

💬 GAIO Squad Riya, Kabir, Bolt • online

👧 Riya
Bolt, a classmate submitted an AI-written essay as his own 📝.
10:02

🤖 Bolt
That's plagiarism 🚫: presenting someone else's work, including AI's, as yours.
10:04 ✓✓

👦 Kabir
But what if we just use AI for ideas?
10:06

🤖 Bolt
Then disclose it 🏷️: 'AI used for brainstorming.' Follow your school's rules.
10:08 ✓✓

👧 Riya
Can teachers tell AI-written text?
10:10

🤖 Bolt
Detectors aren't reliable ⚠️. But some AI tools add hidden watermarks, and teachers know your voice.
10:12 ✓✓

Kabir
How do we keep work original?
10:14

Bolt
Add your own thinking, examples and experiences 💭, and cite all sources.
10:16 ✓✓

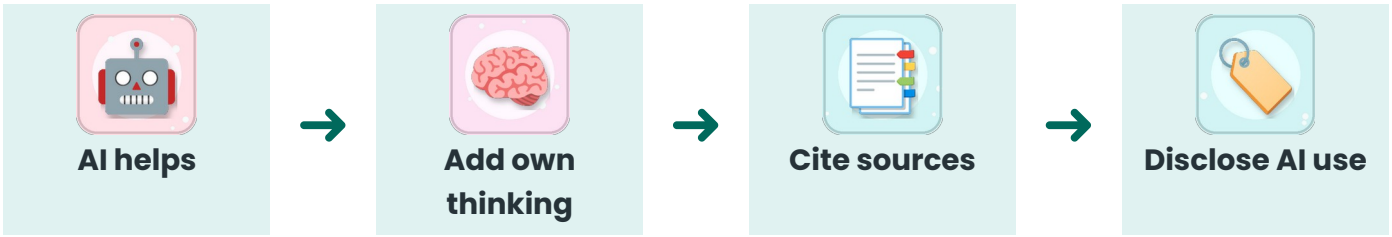
Riya
Disclose, cite, think for yourself ✅!
10:18

😊 Type a message...

📌 **Pinned by Bolt**

Passing off AI work as your own is plagiarism. Disclose AI use, cite sources, follow school rules and add your own thinking.

🔄 How it works

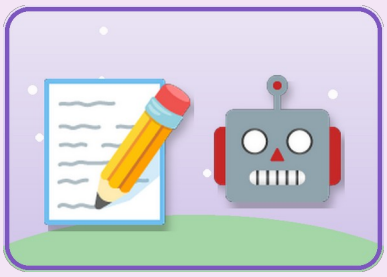


INDIA SPOTLIGHT

Indian universities and boards are framing guidelines on acceptable AI use in assignments and examinations.

AI CHRONICLES: The Essay That Wasn't His

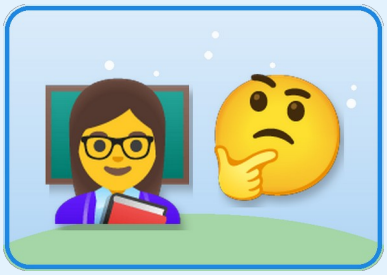
Arun Riya Ms. Khan



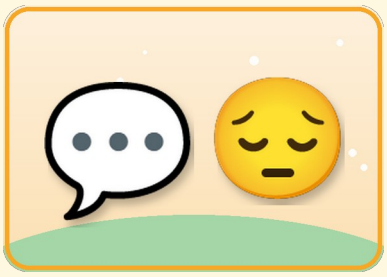
SCENE 1

Arun submitted a brilliant essay on climate change. It was entirely AI-written 😶.

Ms. Khan noticed it didn't sound like Arun, and it had no personal examples or sources.



SCENE 2



SCENE 3

When she asked about one paragraph, Arun couldn't explain it. He admitted the truth.

Ms. Khan explained: "Using AI isn't the problem. Hiding it is plagiarism."

SCENE 4

SCENE 5

Riya showed her essay: own ideas, cited sources and a note: 'AI used for grammar checks.'

Arun rewrote his essay honestly with his village's flood story. His best work yet 🌟.

SCENE 6

🔮 What happens next? Tick ✓ one!

☐	☐	☐
Arun helps write the class AI policy	**Arun hides AI use again**	**The essay writes an apology**

🧐 Story check: answer the questions.

1. How did Ms. Khan notice a problem?

2. What is plagiarism?

3. How did Riya disclose AI use?

Rate this story: shade the stars.

☆ ☆ ☆ ☆ ☆

LOOK AROUND YOU: Where originality matters

School essays

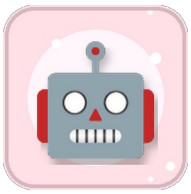
Art competitions

Science projects

Journalism

Music releases

College applications



BOLT SAYS:

Using AI isn't cheating if allowed and disclosed. Hiding it is!

FUN FACT!

Some AI companies add invisible watermarks to AI-made images to help identify them!

I spy AI! Draw one AI helper you saw today.

LET'S DO IT! Human or AI? Debate

You need:

Sample texts

Voting cards

Teams

Steps:

1		Teacher shares 4 short paragraphs.
2		Teams guess: human-written or AI-written?
3		Vote and explain clues.
4		Debate: when is using AI fair in school?

THINK & TALK

Was it easy to tell? What does that mean for honesty?

Draw what you did!

WORKSHEET: Original or AI-Made? (Part A)

Name: _____ Date: _____

1. Honest or plagiarism? Tick ✓.

	Honest	Plagiarism
Essay with "AI helped brainstorm" note		
AI essay submitted as mine		
Quoting with citation		
AI art entered as hand-drawn		
My ideas + AI grammar check (allowed)		

2. Fill in the blanks. Use the word box.

WORD BOX: watermark ✿ plagiarism ✿ disclosure

Using others' work as yours is _____.

Saying AI helped is _____.

A hidden ID mark is a _____.

WORKSHEET: Original or AI-Made? (Part B)

Name: _____ Date: _____

1. Match the term to its meaning.

Plagiarism ●

Disclosure ●

Watermark ●

Citation ●

● Credit to a source

● Hidden AI mark

● Copying as your own

● Saying AI helped

2. Write your own "AI use statement" for a school project!

AI helped me by: _____

PUZZLE FUN: Original or AI-Made?

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.

A				
B				
C				

2. Define it! Write each term in your own words with an example.

Plagiarism

Meaning: _____

Example: _____

Disclosure

Meaning: _____

Example: _____

Watermark

Meaning: _____

Example: _____

🧪 DEMO LAB: Two texts + disclosure template (with teacher)

Name: _____ Date: _____

🎯 **Goal:** Practise spotting plagiarism and writing honest disclosures.

🔗 **Open:** teacher-prepared samples | 👩‍🏫 **With a teacher or parent**

👣 Steps

1. Read 3 sample answers: original, copied from a website, AI-generated.
2. Discuss which is which and why. Note clues.
3. Run them through an AI detector (if available). Was it right?
4. Write an AI disclosure statement for a past project.
5. Write a correct citation for one website source.

🔬 Record your observations

Detector correct? Yes/No:	
My disclosure statement:	
My citation:	

💭 **Reflect:** Why are AI detectors not enough to judge originality?

? QUIZ TIME: Original or AI-Made?

Circle the right answer!

Q1. Passing off AI work as yours is...

a) Plagiarism

b) Winning

c) Art

Q2. Saying AI helped is...

a) Hiding

b) Disclosure

c) Singing

Q3. A hidden ID mark is a...

a) Wave

b) Balloon

c) Watermark

Q4. Using AI when allowed & disclosed is...

a) Fair

b) Cheating

c) Secret

Q5. Originality needs your own...

a) Pizza

b) Ideas

c) Socks

MY SCORE: ____ / 5. Shade one star per correct answer.

☆ ☆ ☆ ☆ ☆

✅ TRUE OR FALSE: Original or AI-Made?

Look at the picture. Tick ✓ TRUE or FALSE.

		👍 TRUE	👎 FALSE
	Submitting AI work as yours is plagiarism.		
	Disclosing AI use is honest.		
	Hiding AI use is always fine.		
	Some AI images carry hidden watermarks.		
	Adding your own ideas shows originality.		
	AI art can be entered as hand-drawn.		

HOME CONNECT

Talk with your family: when is it fair to use AI for school work?

🔢 Self-assessment: tick one.

I know it well!

I know some.

I need practice.

EXERCISE: Original or AI-Made?

Name: _____ Date: _____

A. Tick (✓) the correct answer.

1. Presenting others' or AI's work as your own is ____.

(a) plagiarism ☐ **(b)** paraphrasing ☐ **(c)** publishing ☐

2. Stating that you used AI is ____.

(a) hiding ☐ **(b)** disclosure ☐ **(c)** deletion ☐

3. A hidden mark some AI tools add to content is a ____.

(a) watchword ☐ **(b)** wallpaper ☐ **(c)** watermark ☐

4. AI detectors are ____.

(a) not fully reliable ☐ **(b)** always correct ☐ **(c)** illegal ☐

B. Fill in the blanks. Use the word box.

cite examples rules hiding

1. Disclose, ____, think for yourself.

2. Add personal ____ and experiences.

3. Follow your school's AI ____.

4. Using AI isn't wrong; ____ it is.

C. Write T for True and F for False.

1. Submitting AI essays as your own is plagiarism. [ ]

2. Disclosing AI use is dishonest. [ ]

3. AI detectors can make mistakes. [ ]

4. Citing sources is part of honest work. [ ]

D. Match the following. Write the letter.

Column A		Column B
1. Plagiarism	____	**(a)** Hidden content mark
2. Disclosure	____	**(b)** Passing off others' work
3. Watermark	____	**(c)** Credit to source
4. Citation	____	**(d)** Admitting AI use

E. Answer in one word.

1. What is passing off AI work as yours? _____

2. What is stating AI use? _____

3. What hidden mark identifies AI content? _____

F. Answer these questions.

1. Write an AI disclosure statement for a science project.

2. How can you keep AI-assisted work original?

G. Case-based questions. Read and answer.

Arun submitted an AI-written essay without disclosure and couldn't explain it. Riya's essay had her own ideas, citations and a note: 'AI used for grammar checks.'

1. Arun's action was _____.

(a) plagiarism ☐ **(b)** disclosure ☐ **(c)** citation ☐

2. Riya's note is a _____.

(a) watermark ☐ **(b)** disclosure ☐ **(c)** detector ☐

3. Arun's rewrite improved by adding _____.

(a) more AI text ☐ **(b)** copied content ☐ **(c)** his own experience ☐

H. THINK DEEPER

Draft a fair AI-use policy for Class 10 assignments: what's allowed, what's not and how to disclose.

TOPIC 12 • MONTH 6 • RESPONSIBLE AI

Use AI, Don't Misuse AI

LET'S LEARN

- AI is powerful, so it must be used responsibly.
- Misuse includes cheating, deepfakes, scams and bullying.
- Laws and school rules apply to AI misuse too!

Responsible AI rules

Be honest

Be safe

Be kind

Follow laws

MAGIC WORDS

Responsible
Using wisely

Misuse Using to harm

Cyberbullying
Online harassment

💬 TECH TALK: Use AI, Don't Misuse AI

💬 GAIO Squad Riya, Kabir, Bolt • online

👦 Kabir
Someone made an AI meme mocking a classmate's face 😠.
10:02

🤖 Bolt
That's cyberbullying 🚫, and AI makes it more harmful. It can deeply hurt someone.
10:04 ✓✓

👧 Riya
What other misuse happens?
10:06

🤖 Bolt
Cheating in exams 📝, deepfakes, scams 💸, hacking help and spreading lies.
10:08 ✓✓

👦 Kabir
Is it illegal?
10:10

🤖 Bolt
Many acts are. Laws and school rules apply to AI misuse too ⚖️. 'The AI did it' is no excuse.
10:12 ✓✓

👧 **Riya**

What should we do if we see it?

10:14

🤖 **Bolt**

Don't share 🛑. Support the victim 🤝, tell a trusted adult and report on the platform or cybercrime.gov.in.

10:16 ✓✓

👦 **Kabir**

Use AI responsibly, stand up for others 💪!

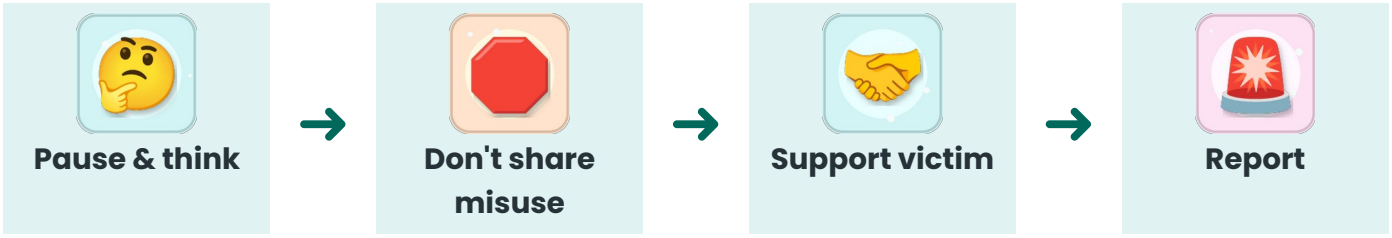
10:18

😊 Type a message...

📌 **Pinned by Bolt**

AI must be used responsibly. Misuse includes cheating, deepfakes, scams and cyberbullying. Laws and rules apply; don't share, support victims and report.

🔄 How it works

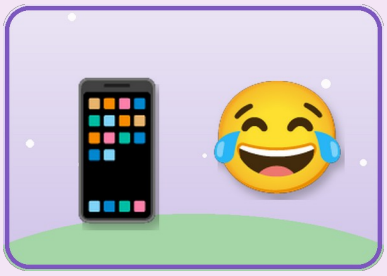


INDIA SPOTLIGHT

India's IT laws cover many cyber offences, and citizens can report cybercrime at cybercrime.gov.in or call 1930.

AI CHRONICLES: The Meme That Hurt

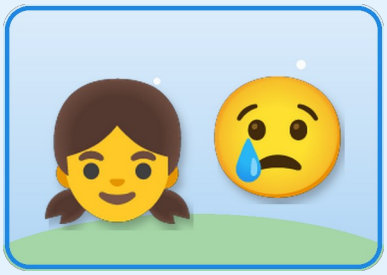
Riya Kabir Sana



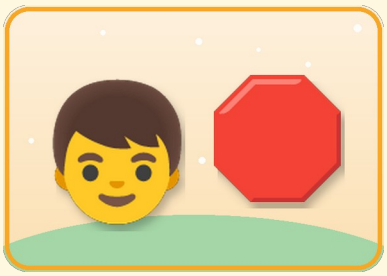
SCENE 1

An AI-edited meme of Sana's face went viral in the class group. Many laughed and forwarded it.

Sana stopped coming to school. She felt humiliated 💔.



SCENE 2



SCENE 3

Kabir refused to forward it and said in the group: "This is cyberbullying. Delete it."

Riya called Sana to support her and went with her to tell the class teacher.

SCENE 4

SCENE 5

The school acted under its anti-bullying policy and the creator faced consequences.

Sana returned. The class made a pledge: "Use AI to lift people up, not to tear them down."

SCENE 6

🔮 What happens next? Tick ✓ one!

☐	☐	☐
The class runs an anti-cyberbullying campaign	**They forward the next meme**	**The meme deletes itself**

🧐 Story check: answer the questions.

1. Why was the meme harmful?

2. What did Kabir and Riya each do?

3. What pledge did the class make?

Rate this story: shade the stars.

☆ ☆ ☆ ☆ ☆

LOOK AROUND YOU: Use vs misuse

Use: study help

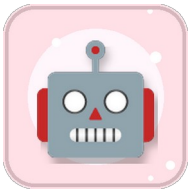
Misuse: cheating in exams

Use: accessibility tools

Misuse: fake images of others

Use: creative projects

Misuse: scam messages



BOLT SAYS:

The same AI can help or harm. YOU decide how it's used!

FUN FACT!

Creating or sharing harmful fake images of a person can be a crime under Indian law!

I spy AI! Draw one AI helper you saw today.

LET'S DO IT! AI Ethics Courtroom

You need:

Case cards

Judge's hammer

Roles

Steps:

1

Read a case: "A student made a fake video of a teacher."

2

Roles: judge, prosecutor, defender, jury.

3

Argue: what harm happened? What rules were broken?

4

Jury decides a fair consequence and a lesson.

THINK & TALK

How should schools handle AI misuse fairly?

Draw what you did!

WORKSHEET: Use AI, Don't Misuse AI (Part A)

Name: _____ Date: _____

1. Responsible use or misuse? Tick ✓.

	Use	Misuse
Revising with AI quizzes		
Using AI in a board exam		
Reading app for a blind friend		
Fake photo of a classmate		
AI voice to trick grandparents		

2. Design a "Responsible AI" pledge poster for your school!

Our pledge: _____

WORKSHEET: Use AI, Don't Misuse AI (Part B)

Name: _____ Date: _____

1. Circle the responsible uses of AI.

Study quizzes

Fake photos of people

Accessibility apps

Scam calls

Creative projects

Translation help

2. Fill in the blanks. Use the word box.

WORD BOX: misuse ✿ responsible ✿ cyberbullying

Using AI wisely is _____.

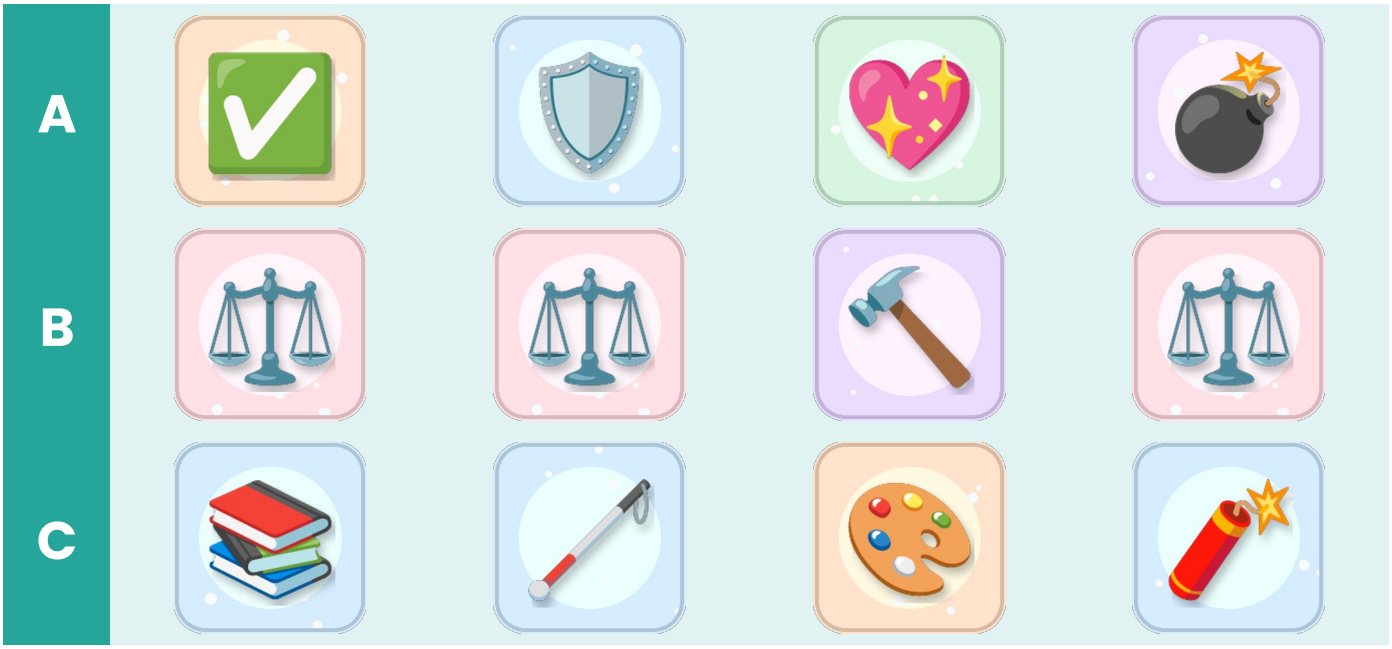
Using AI to harm is _____.

Online harassment is _____.

PUZZLE FUN: Use AI, Don't Misuse AI

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.



2. Define it! Write each term in your own words with an example.

Responsible

Meaning: _____

Example: _____

Misuse

Meaning: _____

Example: _____

Cyberbullying

Meaning: _____

Example: _____

🧪 DEMO LAB: Scenario cards + school AI policy draft (unplugged)

Name: _____ Date: _____

🎯 **Goal:** Decide responsible responses to AI misuse and draft rules.

🔗 **Open:** printed scenario cards | 👩‍🏫 **With a teacher or parent**

👣 Steps

1. Read 6 scenarios: exam cheating, deepfake meme, voice scam, fake review, hacking request, rumour.
2. For each: is it misuse? Who is harmed?
3. Write the right response: don't share, support, report.
4. Find the national cybercrime reporting steps.
5. Draft 5 rules for responsible AI use in your school.

🔬 Record your observations

Most harmful scenario:	
Reporting steps:	
One rule I wrote:	

💭 **Reflect:** Why is 'the AI did it' never an excuse?

? QUIZ TIME: Use AI, Don't Misuse AI

Circle the right answer!

Q1. Using AI to harm is...

a) Misuse	b) Good use	c) Fun

Q2. Online harassment is...

a) Gaming	b) Cyberbullying	c) Music

Q3. Who decides how AI is used?

a) Rain	b) The cat	c) We do

Q4. Fake harmful images of others can be...

a) Illegal	b) Fine	c) Funny

Q5. Using AI in a board exam is...

a) Allowed	b) Cheating	c) Smart

MY SCORE: ____ / 5. Shade one star per correct answer.

☆ ☆ ☆ ☆ ☆

✅ TRUE OR FALSE: Use AI, Don't Misuse AI

Look at the picture. Tick ✓ TRUE or FALSE.

		👍 TRUE	👎 FALSE
	AI misuse can break laws.		
	Fake images of classmates are harmless fun.		
	We decide how AI is used.		
	Cyberbullying is online harassment.		
	Using AI in board exams is allowed.		
	Use AI to help, not harm.		

HOME CONNECT

Write a family "Responsible AI" pledge together and sign it!

🔢 Self-assessment: tick one.

I know it well!

I know some.

I need practice.

EXERCISE: Use AI, Don't Misuse AI

Name: _____ Date: _____

A. Tick (✓) the correct answer.

1. Using AI to mock or harass someone is ____.

(a) cyberbullying ☐ **(b)** cybersecurity ☐ **(c)** cyber art ☐

2. Using AI to cheat in exams is AI ____.

(a) research ☐ **(b)** misuse ☐ **(c)** practice ☐

3. If you see misuse, you should ____.

(a) forward it ☐ **(b)** laugh ☐ **(c)** report it ☐

4. Laws ____ to AI misuse.

(a) apply ☐ **(b)** never apply ☐ **(c)** are unknown ☐

B. Fill in the blanks. Use the word box.

excuse victim cybercrime up

1. 'The AI did it' is no ____.

2. Support the ____.

3. Report at ____.gov.in.

4. Use AI to lift people ____.

C. Write T for True and F for False.

1. Forwarding a hurtful meme spreads harm. [ ]

2. AI misuse is never illegal. [ ]

3. Telling a trusted adult helps. [ ]

4. Deepfake scams are AI misuse. [ ]

D. Match the following. Write the letter.

Column A		Column B
1. Cyberbullying	____	**(a)** Tell authorities/platform
2. Misuse	____	**(b)** Online harassment
3. Report	____	**(c)** Safe, fair, honest
4. Responsible use	____	**(d)** Harmful use

E. Answer in one word.

1. What is online harassment called? _____

2. Which helpline reports cybercrime? _____

3. Should you forward harmful memes? _____

F. Answer these questions.

1. List four forms of AI misuse.

2. What should a bystander do when they see cyberbullying?

G. Case-based questions. Read and answer.

An AI-edited meme of Sana spread in a class group. Kabir refused to forward it and asked for deletion; Riya supported Sana and informed a teacher. The school took action.

1. The meme was ____.

(a) cyberbullying ☐ **(b)** a news report ☐ **(c)** a joke with no harm ☐

2. Kabir acted as an upstander by ____.

(a) forwarding it ☐ **(b)** refusing and speaking up ☐ **(c)** laughing ☐

3. Riya's support helped the ____.

(a) bully ☐ **(b)** meme ☐ **(c)** victim ☐

H. THINK DEEPER

Write a 5-point 'Responsible AI Pledge' for your school and explain why each point matters.

UNIT REVIEW: AI CARE

Name: _____ Date: _____

Circle the right answer!

Q1. Passing off AI work as yours:

a) Plagiarism | b) Winning | c) Art

Q2. Using AI to harm is...

a) Good use | b) Misuse | c) Fun

Q3. Saying AI helped is...

a) Hiding | b) Singing | c) Disclosure

Q4. Online harassment is...

a) Cyberbullying | b) Gaming | c) Music

Q5. A hidden AI ID mark is a...

a) Wave | b) Watermark | c) Balloon

Q6. AI in board exams is...

a) Allowed | b) Smart | c) Cheating

WORD SEARCH: AI CARE

Find these words. They go across → or down ↓.

PLAGIARISM

DISCLOSURE

WATERMARK

MISUSE

ETHICS

HONEST

D	I	P	D	G	K	D	P	B	U
I	G	M	R	G	W	S	L	U	S
S	A	M	H	W	A	H	A	O	E
C	K	M	U	D	T	O	G	M	T
L	C	S	S	G	E	N	I	I	H
O	D	W	I	K	R	E	A	S	I
S	G	N	R	H	M	S	R	U	C
U	E	W	G	S	A	T	I	S	S
R	A	G	F	A	R	L	S	E	T
E	W	W	K	M	K	N	M	S	N

I finished AI CARE! Teacher's sign: _____

MY PICTURE DICTIONARY (1/2)

Augmentation
AI helping humans

Automation
Machines doing tasks

Certificate
Short course proof

Code review
Checking code

Comment
Notes in code

Cyberbullying
Online harassment

Degree
University qualification

Deploy
Put into real use

Disclosure
Saying AI helped

Edge case
Unusual input

Empathy
Feeling with others

Entrance exam
Admission test

Evaluate
Measure performance

Exception
Error during running

Feature
An input property

Hallucination
Confident wrong answer

Ideate
Generate ideas

Industry
A field of business

MY PICTURE DICTIONARY (2/2)

Internship
Short work training

LLM
Large Language Model

Misuse
Using to harm

Networking
Building connections

Overfitting
Memorising, not learning

Pair programming
Coding together

Plagiarism
Using others' work as yours

Pomodoro
Focus-break method

Portfolio
Showcase of work

Priority
Most important task

Productivity
Getting more done well

Prototype
First test model

Responsible
Using wisely

Scoping
Defining the problem

Token
A word piece

Unit test
Check one small part

Watermark
Hidden ID mark

Weight
Importance of a feature

? OLYMPIAD PRACTICE TEST • Part 1 of 4

Read each question. Circle the right answer. 20 questions in all!

Q1. For predicting house price, "size" is a...

a) Feature

b) Label

c) Bug

Q2. An AI that scores 100% on old data but fails new data is...

a) Perfect

b) Overfitted

c) Sleeping

Q3. "Good ___" → "morning" shows an LLM...

a) Painting

b) Cooking

c) Predicting next token

Q4. A chatbot inventing a fake fact is...

a) Hallucinating

b) Verifying

c) Singing

Q5. print(f"Sum: {2+3}") shows...

a) Error

b) Sum: 5

c) Sum: 2+3

? OLYMPIAD PRACTICE TEST • Part 2 of 4

Q6. You get AI code you don't understand. You...

a) Submit it

b) Ignore errors

c) Ask it to explain

Q7. divide(5, 0) without try/except will...

a) Crash

b) Return 0

c) Celebrate

Q8. assert add(2, 2) == 4 is a...

a) Balloon

b) Unit test

c) Bug

Q9. Comforting a patient needs...

a) Automation

b) Only AI

c) Humans

Q10. Design thinking step after Empathise:

a) Define

b) Deploy

c) Delete

? OLYMPIAD PRACTICE TEST • Part 3 of 4

Q11. Predicting canteen diners helps reduce...

a) Music

b) Food waste

c) Books

Q12. A course focusing on statistics & data:

a) B.Des

b) Hotel management

c) B.Sc Data Science

Q13. Debate & MUN build...

a) Communication

b) Cooking

c) Swimming

Q14. Pomodoro breaks are usually about...

a) 2 hours

b) 5 minutes

c) 2 days

Q15. Training a fruit model needs photos of...

a) Cars

b) Dogs

c) Fruits

? OLYMPIAD PRACTICE TEST • Part 4 of 4

Q16. AI project cycle: Problem → Data → ...

a) Model

b) Cake

c) Sleep

Q17. "AI helped me brainstorm" is...

a) Plagiarism

b) Disclosure

c) Hiding

Q18. Entering AI art as hand-drawn is...

a) Honest

b) Smart

c) Dishonest

Q19. Using AI voice to trick grandparents is...

a) Misuse

b) Good use

c) Funny

Q20. The same AI can help or harm. Who decides?

a) Weather

b) Humans

c) Luck

MY SCORE: I got _____ / 20. Colour a star for every 4 right!

CERTIFICATE

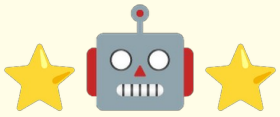
of AI Explorer

This is to say that

_____

of Class 10 finished the 6-month AI journey:

DISCOVER → CONNECT → SOLVE → RISE → CREATE → CARE



Teacher: _____ Date: _____

? ANSWER KEY (for teachers & parents)

1. How AI Thinks with Data	**Quiz:** 1-a 2-b 3-c 4-a 5-b **True/False:** 1-T 2-T 3-F 4-T 5-T 6-F **Odd one out:** A: picture 4; B: picture 3; C: picture 4 **Part A 1:** Feature - Input property; Weight - Importance of input; Label - Correct answer tag; Overfitting - Memorising data **Part A 2:** Size in sq ft: Feature; Number of rooms: Feature; House price: Label; Location: Feature **Part B 1:** a) Feature, c) 3, b) Up **Part B 2:** feature, weight, overfitting **Exercise A:** 1-a 2-b 3-c 4-a **Exercise B:** error, weight, new, overfitting **Exercise C:** 1-T 2-F 3-T 4-F **Exercise D:** 1-b 2-d 3-a 4-c **Exercise E:** Features; Weight; Overfitting **Story next:** Any sensible choice; best: They add a 'season' feature and retest **Exercise G (case):** 1-a 2-b 3-c **Think deeper:** Open answer; reward reasoning and examples.
2. Generative AI Uncovered	**Quiz:** 1-a 2-b 3-c 4-a 5-b **True/False:** 1-T 2-T 3-F 4-T 5-T 6-F **Odd one out:** A: picture 4; B: picture 3; C: picture 4 **Part A 1:** morning, tea, day **Part A 2:** LLMs predict next words: True; LLMs truly understand like humans: False; GenAI can hallucinate: True; GenAI is always correct: False **Part B 1:** Chatbot - Text answers; Image AI - Pictures; Music AI - Tunes; Code AI - Programs **Part B 2:** LLM, token, hallucination **Exercise A:** 1-a 2-b 3-c 4-a **Exercise B:** Language, piece, verified, patterns **Exercise C:** 1-T 2-F 3-T 4-F **Exercise D:** 1-b 2-d 3-a 4-c **Exercise E:** Token; Hallucination; Noise **Story next:** Any sensible choice; best: Kabir teaches classmates to fact-check AI **Exercise G (case):** 1-a 2-b 3-c **Think deeper:** Open answer; reward reasoning and examples.

3. AI-Assisted Coding	**Quiz:** 1-a 2-b 3-c 4-a 5-b **True/False:** 1-T 2-F 3-T 4-T 5-F 6-T **Odd one out:** A: picture 4; B: picture 3; C: picture 4 **Part A 1:** Ask AI to explain an error: Good; Submit AI code you can't explain: Bad; Ask AI for test ideas: Good; Skip testing AI code: Bad **Part A 2:** Hi Meera, 15, Pass **Part B 1:** Explain error - Simple explanation; Suggest tests - Test ideas; Add comments - Notes in code; Clean code - Neater code **Part B 2:** Child's own work **Exercise A:** 1-a 2-b 3-c 4-a **Exercise B:** understand, explain, bugs, hashed **Exercise C:** 1-T 2-F 3-T 4-T **Exercise D:** 1-b 2-d 3-a 4-c **Exercise E:** Comment; Code review; Pair **Story next:** Any sensible choice; best: They add two-factor login **Exercise G (case):** 1-a 2-b 3-c **Think deeper:** Open answer; reward reasoning and examples.
4. Code → Test → Improve	**Quiz:** 1-a 2-b 3-c 4-a 5-b **True/False:** 1-T 2-T 3-F 4-T 5-F 6-T **Odd one out:** A: picture 4; B: picture 3; C: picture 4 **Part A 1:** 1-Write code, 2-Test, 3-Fix bugs, 4-Improve **Part A 2:** marks = 0: Edge case; marks = 55: Normal; marks = 100: Edge case; marks = -5: Edge case; marks = 72: Normal **Part B 1:** try, except, assert **Part B 2:** Unit test - Check one part; Edge case - Unusual input; try/except - Handle errors; Iterate - Improve again & again **Exercise A:** 1-a 2-b 3-c 4-a **Exercise B:** Value, condition, negative, repeat **Exercise C:** 1-T 2-F 3-T 4-F **Exercise D:** 1-b 2-d 3-a 4-c **Exercise E:** try/except; Unit test; ValueError **Story next:** Any sensible choice; best: They write tests first next time **Exercise G (case):** 1-a 2-b 3-c **Think deeper:** Open answer; reward reasoning and examples.
5. AI at Work	**Quiz:** 1-a 2-b 3-c 4-a 5-b **True/False:** 1-T 2-T 3-F 4-T 5-T 6-F **Odd one out:** A: picture 4; B: picture 3; C: picture 4 **Part A 1:** Bank - Fraud detection; Factory - Defect spotting; Delivery - Best routes; Shop - Stock prediction **Part A 2:** Sorting thousands of files: Automate; Comforting a patient: Human; Checking bill totals: Automate; A judge's final decision: Human **Part B 1:** Circle: Sorting files, Checking totals, Filtering spam, Spotting defects **Part B 2:** automation, augmentation, industry **Exercise A:** 1-a 2-b 3-c 4-a **Exercise B:** decide, automate, with, human **Exercise C:** 1-T 2-F 3-T 4-F **Exercise D:** 1-b 2-d 3-a 4-c **Exercise E:** Automation; Augmentation; Creativity / care / judgement **Story next:** Any sensible choice; best: Uncle Ravi trains other officers **Exercise G (case):** 1-a 2-b 3-c **Think deeper:** Open answer; reward reasoning and examples.

6. AI Solving Real Problems

Quiz: 1-a 2-b 3-c 4-a 5-b
True/False: 1-T 2-F 3-T 4-T 5-T 6-F
Odd one out: A: picture 4; B: picture 3; C: picture 4
Part A 1: 1-Empathise, 2-Define, 3-Ideate, 4-Prototype & test
Part A 2: Leaks - Sensor alerts; Food waste - Demand forecast; Cattle - Photo diagnosis; Reading - Reading tutor
Part B 1: Empathise - Interview users; Define - Write one problem sentence; Ideate - Brainstorm 10 ideas; Prototype - Build a quick model
Part B 2: Child's own work
Exercise A: 1-a 2-b 3-c 4-a
Exercise B: test, user, paper, tool
Exercise C: 1-T 2-F 3-T 4-F
Exercise D: 1-b 2-d 3-a 4-c
Exercise E: Empathise; Prototype; Test
Story next: Any sensible choice; best: They make it for 10 more elders
Exercise G (case): 1-a 2-b 3-c
Think deeper: Open answer; reward reasoning and examples.

7. Your Road to an AI Career

Quiz: 1-a 2-b 3-c 4-a 5-b
True/False: 1-T 2-F 3-T 4-T 5-T 6-F
Odd one out: A: picture 4; B: picture 3; C: picture 4
Part A 1: B.Tech - Building AI systems; B.Sc DS - Statistics & data; B.Des - User experience; BBA - Business data
Part A 2: degree, certificate, entrance exam
Part B 1: B.Tech CS: Yes; B.Sc Data Science: Yes; B.Des UX: Yes; Online AI certificate: Yes
Part B 2: Child's own work
Exercise A: 1-a 2-b 3-c 4-a
Exercise B: goals, degrees, analytics, courses
Exercise C: 1-T 2-F 3-T 4-T
Exercise D: 1-b 2-d 3-a 4-c
Exercise E: Entrance exam; B.Tech AI / B.Sc Data Science; Business analytics
Story next: Any sensible choice; best: They co-create a project combining both skills
Exercise G (case): 1-a 2-b 3-c
Think deeper: Open answer; reward reasoning and examples.

8. AI Skills Beyond School

Quiz: 1-a 2-b 3-c 4-a 5-b
True/False: 1-T 2-T 3-F 4-T 5-F 6-T
Odd one out: A: picture 4; B: picture 3; C: picture 4
Part A 1: Circle: Online course, Hackathon, Personal project, Debate club
Part A 2: portfolio, networking, internship
Part B 1: Debate - Speaking; Hackathon - Problem solving; Volunteering - Empathy; Online course - Technical knowledge
Part B 2: Child's own answer
Exercise A: 1-a 2-b 3-c 4-a
Exercise B: skills, communication, competitions, code
Exercise C: 1-T 2-F 3-T 4-F
Exercise D: 1-b 2-d 3-a 4-c
Exercise E: Portfolio; Internship; SWAYAM / NPTEL
Story next: Any sensible choice; best: Kabir mentors juniors next summer
Exercise G (case): 1-a 2-b 3-c
Think deeper: Open answer; reward reasoning and examples.

9. AI Productivity Booster

Quiz: 1-a 2-b 3-c 4-a 5-b
True/False: 1-T 2-T 3-F 4-T 5-F 6-T
Odd one out: A: picture 4; B: picture 3; C: picture 4
Part A 1: 1-Choose top priority, 2-Focus 25 minutes, 3-Take a short break, 4-Review & summarise
Part A 2: 25-min focus sessions: Productive; Phone notifications on: Not; Weekly plan: Productive; Studying all night daily: Not; Ticking a to-do list: Productive
Part B 1: Planner - Weekly schedule; Timer - Focus sessions; To-do list - Tracking tasks; Summariser - Quick revision
Part B 2: Pomodoro, priority, productivity
Exercise A: 1-a 2-b 3-c 4-a
Exercise B: important, summarise, away, sleep
Exercise C: 1-T 2-F 3-T 4-F
Exercise D: 1-b 2-d 3-a 4-c
Exercise E: Pomodoro; Priorities; Textbook
Story next: Any sensible choice; best: Riya shares her planner with friends
Exercise G (case): 1-a 2-b 3-c
Think deeper: Open answer; reward reasoning and examples.

10. Create Your AI Project

Quiz: 1-a 2-b 3-c 4-a 5-b
True/False: 1-T 2-T 3-F 4-T 5-T 6-F
Odd one out: A: picture 4; B: picture 3; C: picture 4
Part A 1: 1-Problem scoping, 2-Collect data, 3-Build model, 4-Deploy
Part A 2: Child's own work
Part B 1: Scoping - Define the problem; Data - Collect photos; Evaluate - Measure accuracy; Deploy - Give to real users
Part B 2: scoping, evaluate, deploy
Exercise A: 1-a 2-b 3-c 4-a
Exercise B: deploy, labelled, small, mistakes
Exercise C: 1-T 2-F 3-T 4-F
Exercise D: 1-b 2-d 3-a 4-c
Exercise E: Scoping; Deploying; Documentation
Story next: Any sensible choice; best: They add advice on treatment
Exercise G (case): 1-a 2-b 3-c
Think deeper: Open answer; reward reasoning and examples.

11. Original or AI-Made?

Quiz: 1-a 2-b 3-c 4-a 5-b
True/False: 1-T 2-T 3-F 4-T 5-T 6-F
Odd one out: A: picture 4; B: picture 3; C: picture 4
Part A 1: Essay with "AI helped brainstorm" note: Honest; AI essay submitted as mine: Plagiarism; Quoting with citation: Honest; AI art entered as hand-drawn: Plagiarism; My ideas + AI grammar check (allowed): Honest
Part A 2: plagiarism, disclosure, watermark
Part B 1: Plagiarism - Copying as your own; Disclosure - Saying AI helped; Watermark - Hidden AI mark; Citation - Credit to a source
Part B 2: Child's own work
Exercise A: 1-a 2-b 3-c 4-a
Exercise B: cite, examples, rules, hiding
Exercise C: 1-T 2-F 3-T 4-T
Exercise D: 1-b 2-d 3-a 4-c
Exercise E: Plagiarism; Disclosure; Watermark
Story next: Any sensible choice; best: Arun helps write the class AI policy
Exercise G (case): 1-a 2-b 3-c
Think deeper: Open answer; reward reasoning and examples.

12. Use AI, Don't Misuse AI

Quiz: 1-a 2-b 3-c 4-a 5-b
True/False: 1-T 2-F 3-T 4-T 5-F 6-T
Odd one out: A: picture 4; B: picture 3; C: picture 4
Part A 1: Revising with AI quizzes: Use; Using AI in a board exam: Misuse; Reading app for a blind friend: Use; Fake photo of a classmate: Misuse; AI voice to trick grandparents: Misuse
Part A 2: Child's own work
Part B 1: Circle: Study quizzes, Accessibility apps, Creative projects, Translation help
Part B 2: responsible, misuse, cyberbullying
Exercise A: 1-a 2-b 3-c 4-a
Exercise B: excuse, victim, cybercrime, up
Exercise C: 1-T 2-F 3-T 4-T
Exercise D: 1-b 2-d 3-a 4-c
Exercise E: Cyberbullying; 1930; No
Story next: Any sensible choice; best: The class runs an anti-cyberbullying campaign
Exercise G (case): 1-a 2-b 3-c
Think deeper: Open answer; reward reasoning and examples.

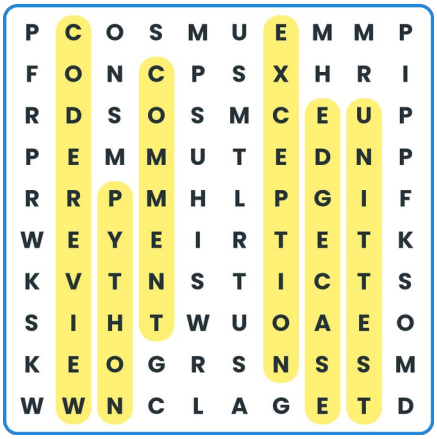
Unit reviews & practice test

AI DISCOVER: 1-a 2-b 3-c 4-a 5-b 6-c • AI CONNECT: 1-a 2-b 3-c 4-a 5-b 6-c • AI SOLVE: 1-a 2-b 3-c 4-a 5-b 6-c • AI RISE: 1-a 2-b 3-c 4-a 5-b 6-c • AI CREATE: 1-a 2-b 3-c 4-a 5-b 6-c • AI CARE: 1-a 2-b 3-c 4-a 5-b 6-c

Olympiad Practice Test: 1-a 2-b 3-c 4-a 5-b 6-c 7-a 8-b 9-c 10-a 11-b 12-c 13-a 14-b 15-c 16-a 17-b 18-c 19-a 20-b

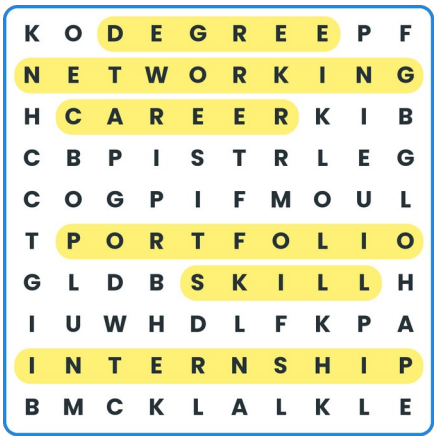
WORD SEARCH ANSWERS

AI DISCOVER

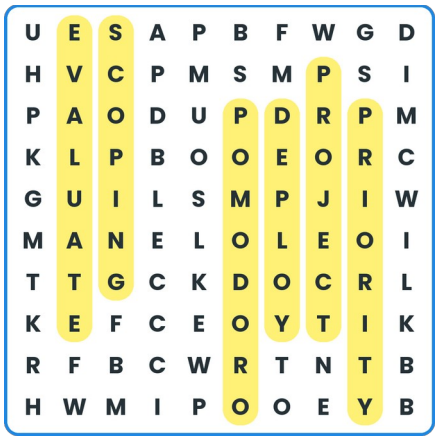


AI CONNECT

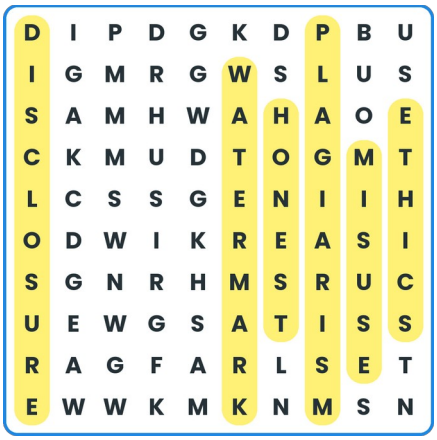
AI SOLVE



AI RISE



AI CREATE



AI CARE

MY NOTES

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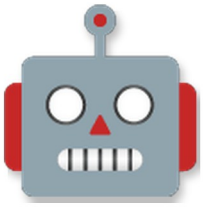

About this book

A hands-on AI skills book for senior learners of Class 10, following the 6-month Global AI Olympiad journey.

- 💬 Tech Talk: concepts as a group chat
- 🧪 Demo Lab: hands-on real AI tools
- 📚 AI Chronicles: 12 illustrated stories
- 📝 Standard, case-based & HOTS questions
- 🏆 Olympiad practice test & certificate

AI Basics | **Communicate with AI** | **AI Applications**

AI & Career | **AI Tools** | **Responsible AI**

Think | Explore | Create | A Better Tomorrow

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